परीक्षा पुस्तिका संकेत
Test Booklet Code

English + Hindi

A

NEET REAL TEST (2025-26)

इस परीक्षा पुस्तिका को तब तक न खोलें जब तक कहा न जाए।
Do not open this Test Booklet until you are asked to do so.
इस परीक्षा पुस्तिका के पिछले आवरण पर दिए निर्देशों को ध्यान से पढ़ें।
Read carefully the Instructions on the Back Cover of this Test Booklet.

Date: 17-08-2025

DURATION: 180 Minutes | **CLASS – Dropper** | **M. MARKS: 300**

Important Instructions:

1. (a) Total duration of NEET UG Paper is **180** min.
   (b) NEET UG Paper Contains **180** question.
   (c) Maximum Marks **720**
2. The question paper consists of 3 Subjects (Physics, Chemistry, Biology).
3. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong response.

महत्वपूर्ण निर्देशः

1. (a) सभी प्रश्नों को हल करने की कुल अवधि NEET UG Paper के लिए **180** मिनट है।
   (b) NEET UG Paper में **180** प्रश्न है।
   (c) अधिकतम अंक **720**
2. इस प्रश्न पत्र में 3 विषय (भौतिक विज्ञान, रसायन विज्ञान, जीव विज्ञान) हैं।
3. प्रत्येक सही उत्तर पर 4 अंक दिए जाएंगे जबकि प्रश्नों के लिए गलत उत्तर के लिए 1 अंक काटा जाएगा।

SYLLABUS

Physics:	**Mathematical Tools, Units and Measurements, Motion in a Straight Line, Vectors and Motion in a Plane**
Chemistry:	**Some Basic Concepts of Chemistry, Redox Reactions, Solutions, Thermochemistry, Thermodynamics, Chemical Equilibrium, Ionic Equilibrium**
Botany:	**Cell - The Unit of Life (Complete Chapter), Cell Cycle and Cell Division (Complete Chapter), The Living World (Complete Chapter), Biological Classification (Complete Chapter)**
Zoology:	**Structural Organisation in Animals, Breathing and Exchange of Gases, Body Fluids and Circulation**

PW Test Series

RealTest_by_PW

PW Test Series

1. A particle is moving along a straight line with velocity $v(t) = (t^2 - 3t + 2)$ m s^{-1}. Find the distance covered by the particle in between $t = 0$ to $t = 3$ s. Given at $t = 0$, $x = 0$.

(1) 2 m (2) $\frac{13}{6}$ m
(3) $\frac{11}{6}$ m (4) $\frac{5}{6}$ m

2. Which of the following statements is/are **correct**?
(1) If a particle is at the origin, its velocity must be zero.
(2) If particle velocity is zero, its acceleration must be zero.
(3) If a particle is at rest, its acceleration may or may not be zero
(4) Both (1) & (3)

3. A particle moves with initial velocity v_0 and retardation αv, where v is the velocity at any time t. Then which of the following statement is **incorrect?**
(1) The particle will cover a total distance (v_0/α) before coming to rest
(2) The velocity of particle at $t = (1/\alpha)$ will be (v_0/e).
(3) The particle will come to rest after a very long time
(4) The particle velocity will be ev_0 after time $t = (1/\alpha)$

4. A particle is moving in a straight line. Its position x at time t is given by $x = \sqrt{1+t^2}$. Its acceleration at time t is:
(1) $\frac{1}{x} - \frac{t^2}{x^3}$
(2) $\frac{1}{x^3}$
(3) $\frac{1}{x} - \frac{t^2}{x^2}$
(4) Both (1) & (2)

5. In a measurement, systematic error in observations is 's'. The systematic error in 100 observations will be:
(1) $\frac{s}{10}$ (2) $10\,s$
(3) s (4) $100\,s$

6. The side of a cube is (2.00 ± 0.01) cm. The surface area of the cube up to **correct** significant figures can be represented by:
(1) (24.00 ± 0.24) cm^2
(2) (24.0 ± 0.2) cm^2
(3) (24.0 ± 0.34) cm^2
(4) (24.0 ± 0.3) cm^2

7. At $t = 0$, a particle is at rest at position $x = 0$. The acceleration of the particle depends on x as $a = \frac{3x^2}{2}$ m s^{-2}. The speed of particle at $t = 2$ s will be:
(1) 4 m s^{-1} (2) 8 m s^{-1}
(3) 2 m s^{-1} (4) zero

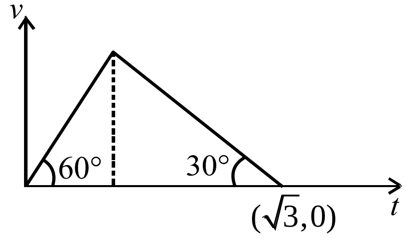
8. A particle is moving in x-y plane with constant acceleration $-g\hat{j}$. The equation of trajectory of particle is given by $y = 2x - 4x^2$. The speed of particle, when it crosses the origin is:
(1) $\sqrt{\frac{8g}{5}}$ (2) $\sqrt{\frac{4g}{5}}$
(3) $\sqrt{\frac{5g}{4}}$ (4) $\sqrt{\frac{5g}{8}}$

9. A cylindrical wire is having length l = (2.00 ± 0.04) unit, radius r = (1.00 ± 0.02) unit and mass m = (2.00 ± 0.04) unit. The maximum percentage error in density will be:
(1) 8% (2) 6%
(3) 10% (4) 4%

10. A particle starts from rest and moves with an acceleration a = 2t m s^{-2}. The velocity of particle at t = 1 s is:
(1) $\frac{1}{2}$ m s^{-1} (2) 2 m s^{-1}
(3) 1 m s^{-1} (4) 4 m s^{-1}

11. A particle is moving in x-y plane with velocity $\vec{v} = k(y\hat{i} + x\hat{j})$. The general equation for its trajectory will be:
(1) $y = x^2$ + constant
(2) $y^2 + x$ = constant
(3) $y^2 = x^2$ + constant
(4) yx = constant

12. In the velocity-time (v-t) graph as shown, two straight lines make angle 60° and 30° with time axis. The maximum velocity attained by the particle is:



(1) $\frac{\sqrt{3}}{4}$ (2) $\frac{3}{4}$

(3) $\frac{3}{2}$ (4) $\sqrt{3}$

13. The velocity v of a particle varies with time t as $v = |t-8|\,\text{m s}^{-1}$. The average velocity of particle between t = 0 to t = 16 s is:

(1) zero (2) 64 m s^{-1}

(3) 4 m s^{-1} (4) 32 m s^{-1}

14. In a hypothetical but physically correct equation, electrical permittivity of a medium (ε) is given by $\varepsilon = a\sqrt{F} + bx^2$, where F is force and x is position. The dimensional formula of $\left(\frac{a}{b}\right)$ is:

(1) $[M^{-\frac{1}{2}}L^{\frac{3}{2}}T]$

(2) $[M^{\frac{1}{2}}L^{-\frac{3}{2}}T^{-1}]$

(3) $[M^{-1}LT^2]$

(4) $[M^2L^{-1}T^2A^{-2}]$

15. **Statement-I:** For a particle moving in a straight line, the average velocity in a time interval is always equals to $\left(\frac{u+v}{2}\right)$, where u and v are initial and final velocity of the particle given time interval.

Statement-II: For a particle moving in a straight line with constant acceleration, the average velocity of the particle cannot be zero in any time interval.

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

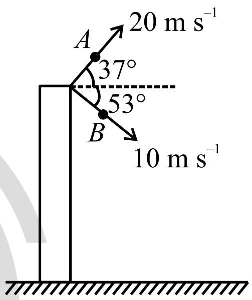
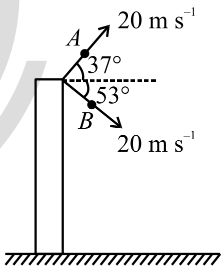
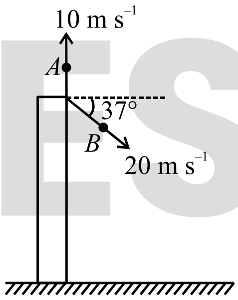
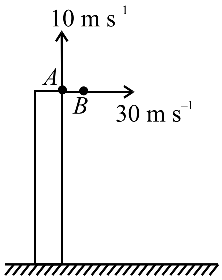
(4) Both Statement I and Statement II are incorrect.

16. A swimmer can swim in still water with a speed of $\sqrt{5}$ m s^{-1}. While crossing the river, his average speed w.r.t. ground is 3 m s^{-1}. If he crosses the river in the shortest possible time, what is the speed of flow of water?

(1) 4 m s^{-1} (2) 2 m s^{-1}

(3) 8 m s^{-1} (4) 6 m s^{-1}

17. List-I gives the figure of two balls A and B thrown simultaneously from a very high tower. Match the entries in List-I with **correct** entries in List-II.

List - I		List - II	
(A)		(I)	Relative velocity of A w.r.t. B is $20\sqrt{2}$ m s^{-1}
(B)		(II)	Relative velocity of A w.r.t. B is $10\sqrt{5}$ m s^{-1}
(C)		(III)	Relative velocity of A w.r.t. B is $10\sqrt{10}$ m s^{-1}
(D)		(IV)	Relative velocity of A w.r.t. B is $\sqrt{740}$ m s^{-1}

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-I, C-IV, D-II

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-I, C-II, D-III

18. A particle is moving in the x-y plane. (x is along the horizontal and y is along the vertical). Its x and y co-ordinates at time t are given by $x = (6t)$ m and $y = (8t - 2t^2)$ m. Its maximum height from the origin is equal to:

(1) 6.4 m (2) 8 m
(3) 9.6 m (4) 32 m

19. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

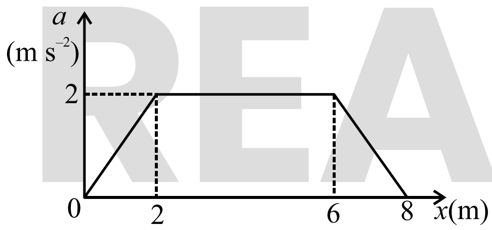
Assertion A: When a swimmer crosses a flowing river, his shortest path is also his quickest path.

Reason R: In this case, as distance covered is minimum in shortest path, so time taken is also minimum.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are correct and R is the correct explanation of A.
(2) Both A and R are correct but R is not the correct explanation of A.
(3) A is correct but R is incorrect.
(4) Both A and R are false.

20. For a particle moving in a straight line, its acceleration-position (a-x) graph is as shown. If velocity of particle at $x = 0$ is 2 m s^{-1}, then its velocity at $x = 8$ m will be:

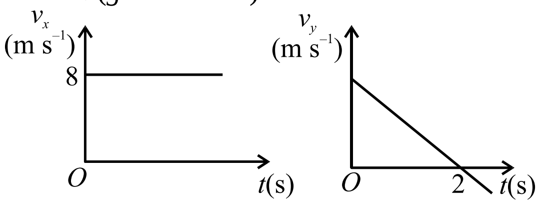


(1) 28 m s^{-1}
(2) 5 m s^{-1}
(3) $2\sqrt{7}$ m s^{-1}
(4) $7\sqrt{2}$ m s^{-1}

21. Evaluate $\int (e^{ax} - e^{-ax})dx$.

(1) $\dfrac{e^{ax} - e^{-ax}}{a} + c$
(2) $\dfrac{e^{ax} + e^{-ax}}{a} + c$
(3) $e^{ax} + e^{-ax} + c$
(4) $e^{ax} - e^{-ax} + c$

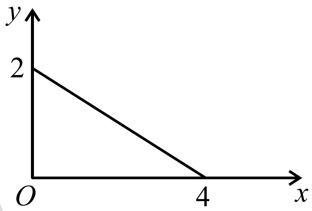
22. A body is projected in a vertical plane (x-y). Its horizontal component of velocity (v_x) and vertical component of velocity (v_y) is plotted against time as shown. (g = 10 m s^{-2})



The horizontal range of the particle is:

(1) 32 m (2) 64 m
(3) 24 m (4) Data insufficient

23. The equation of straight line for the following graph will be:



(1) $x + 2y - 4 = 0$ (2) $x + 2y + 4 = 0$
(3) $2x + y - 4 = 0$ (4) $2x + y + 4 = 0$

24. The ceiling of a long hall is 20 m high. What is the maximum horizontal distance that a ball thrown with a speed 40 m s^{-1} can go without hitting the ceiling of the hall? (Given g = 10 m s^{-2})

(1) 150 m (2) $80\sqrt{3}$ m
(3) 80 m (4) 40 m

25. Two stones are thrown vertically up simultaneously from the edge of a cliff 200 m high with initial speeds of 15 m s^{-1} and 100 m s^{-1} respectively. What is the maximum separation between them during entire motion? (Take g = 10 m s^{-2})

(1) 120 m (2) 700 m
(3) 680 m (4) 850 m

26. Given $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$. Then:

(a) The magnitude of $(\vec{A} + \vec{C})$ equals the magnitude of $(\vec{B} + \vec{D})$
(b) $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$ each must be a null vector
(c) The magnitude of $\vec{A}$ can never be greater than the sum of magnitudes of $\vec{B}, \vec{C}$ and $\vec{D}$.
(d) $(\vec{B} + \vec{C})$ must lie in the plane of $\vec{A}$ and $\vec{D}$, if $\vec{A}$ and $\vec{D}$ are not collinear and in the line of $\vec{A}$ and $\vec{D}$ if they are collinear

The **correct** statement(s) are:

(1) (a), (b), (c) and (d)
(2) (a), (c) and (d) only
(3) (b), (c) and (d) only
(4) (a) and (d) only

27. Given $y = (3 \sin x + 4 \cos x)$. The minimum value of y is:

(1) 5 (2) –5
(3) 7 (4) –7

28. In uniform circular motion, what is not uniform?

(1) speed
(2) velocity
(3) acceleration
(4) both velocity and acceleration

29. In uniform circular motion, acceleration of particle is given by (where symbols have their usual meanings)

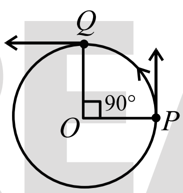
(1) $\vec{a} = \frac{v^2}{r}\hat{r}$

(2) $\vec{a} = -\frac{v^2}{r^3}\vec{r}$

(3) $\vec{a} = \omega^2 \vec{r}$

(4) $\vec{a} = -\frac{v^2 \vec{r}}{r^2}$

30. A particle is moving in a circular path of radius R with constant speed v. When it moves from point P to Q, the value of $\Delta|\vec{v}|$ is:

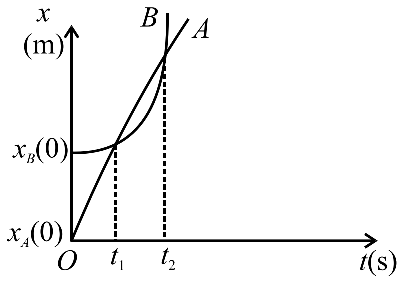


(1) $\sqrt{2}v$
(2) zero
(3) $\sqrt{3}v$
(4) $2\,v$

31. In an experiment for determination of acceleration due to gravity g, time taken for 20 oscillations is measured as (40 ± 1)s. Then percentage error in the time period is:

(1) 5%
(2) 50%
(3) 2.5%
(4) 10%

32. Position-time (x-t) graph of two particles A and B is as shown.



Then we can conclude that:

(1) At $t = t_1$, velocity of both A and B is same.
(2) At $t = t_1$, displacement of both A and B is same.
(3) At $t = t_1$, position of both A and B is same.
(4) At $t = t_2$, velocity of both A and B is same.

33. A particle is moving along a straight line with velocity v as function of position x as $v = \sqrt{81 + 4x}$. Given $t = 0$; $x = 0$. The acceleration of the particle is dependent on x as:

(1) $a \propto x^{\frac{1}{2}}$ (2) $a \propto x^{\frac{3}{2}}$

(3) $a \propto x^0$ (4) $a \propto \frac{1}{\sqrt{x}}$

34. The order of magnitude and number of significant figures in 5.0×10^3 are respectively.

(1) 3, 2 (2) 4, 2
(3) 3, 3 (4) 4, 3

35. The component of $\vec{A} = (3\hat{i} + 4\hat{j} + 5\hat{k})$ in the direction of $\vec{B} = (\hat{i} + \hat{j})$ is:

(1) $\frac{7}{2}(\hat{i} + \hat{j})$ (2) $\frac{7}{\sqrt{2}}(\hat{i} + \hat{j})$

(3) $\frac{5}{2}(\hat{i} + \hat{j})$ (4) $\frac{5}{\sqrt{2}}(\hat{i} + \hat{j})$

36. Three particles A, B and C lie at the corners of an equilateral triangle of side 'a'. They move with constant speed v, such that A is always headed towards B, B towards C and C towards A. The distance covered by each particle, till they meet, is:

(1) $\frac{a}{\sqrt{3}}$ (2) $\sqrt{3}a$

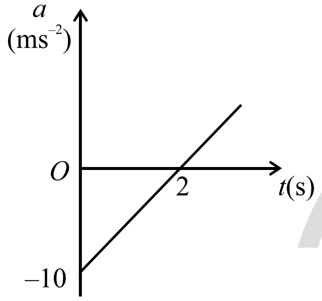
(3) $\frac{2a}{3}$ (4) $3a$

37. A body is projected from ground with speed $u = 10\sqrt{3}\,\text{m s}^{-1}$ at an angle $\theta = 60°$ with the horizontal. At the instant when instantaneous velocity is perpendicular to initial velocity, speed of the body is:

(1) $10\,\text{m s}^{-1}$ (2) $10\sqrt{3}\,\text{m s}^{-1}$

(3) $\frac{10}{\sqrt{3}}\,\text{m s}^{-1}$ (4) $5\,\text{m s}^{-1}$

38. Acceleration-time (a-t) graph of a particle moving in a straight line is as shown. At $t = 0$, velocity v of the particle is -10 m s^{-1}.



Change in speed of particle from $t = 0$ to $t = 2$ s is:

(1) -20 m s^{-1} (2) 10 m s^{-1}

(3) 20 m s^{-1} (4) 30 m s^{-1}

39. In uniform circular motion, which of the following is **correct**? (symbols have their usual meanings)

(1) $\vec{v}.\vec{r} > 0$

(2) $\vec{a}.\vec{r} < 0$

(3) $\vec{a}.\vec{v} = 0$

(4) Both (2) and (3)

40. Two trains A and B, each of length 400 m are moving on two parallel tracks with uniform speed of 20 m s^{-1} in the same direction, A being ahead of B. The driver of B decides to overtake A and accelerates by 1 m s^{-2}. After 50 s, the guard (end point) of B just moves past the driver of A. The original gap between the two trains was:

(1) 450 m (2) 1250 m

(3) 850 m (4) 800 m

41. If $(\vec{A}+\vec{B}+\vec{C}) = \vec{0}$, then:

(1) $\vec{A}\times\vec{B} = \vec{A}\times\vec{C}$

(2) $\vec{A}\times\vec{B} = \vec{C}\times\vec{A}$

(3) $\vec{A}.\vec{B} = \vec{B}.\vec{C}$

(4) $\vec{A}.\vec{C} = \vec{B}.\vec{C}$

42. A bird is flying to and fro between two cars moving towards each other on a straight road. One car has a speed 18 km h^{-1} while the other has a speed of 27 km h^{-1}. The bird starts moving from the first car towards the other with the speed of 36 km h^{-1} and when the two cars were separated by 36 km. Bird instantly reverses her direction after reaching another car and so on. The total distance covered by the bird is:

(1) 28.8 km (2) 36 km

(3) 72 km (4) 42.8 km

43. If $|\vec{A}| = |\vec{B}| = 10$ m and angle between $\vec{A}$ and $-\vec{B}$ is 60° then $|\vec{A}-\vec{B}|$ is:

(1) $10\sqrt{3}$ m (2) 10 m

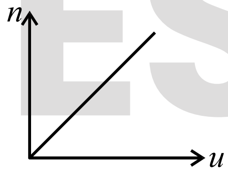
(3) 20 m (4) $5\sqrt{3}$ m

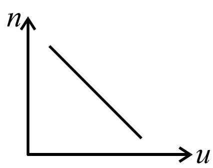
44. An elevator is moving downward with constant acceleration 5 m s^{-2}. A man standing in the elevator drops a ball from a height 2.5 m. The ball reaches the floor of the elevator after a time ($g = 10\text{ m s}^{-2}$)

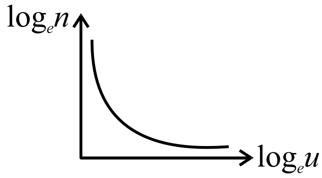
(1) $\frac{1}{\sqrt{2}}$ s (2) 1s

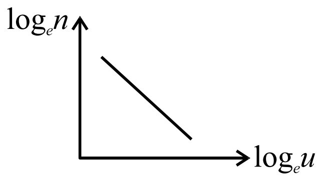
(3) $\sqrt{2}$ s (4) $\frac{1}{\sqrt{3}}$ s

45. In measurements, a certain physical quantity Q is represented in terms of numerical value (n) of quantity and size of unit (u). Which of the following graph correctly represents relation between n and u?

(1)


(2)


(3)


(4)


46. Match **List-I** with **List-II**.

List-I (Redox reactions)		List-II (Type)	
(A)	$3Mg(s) + N_2(g) \rightarrow Mg_3N_2(s)$	(I)	Decomposition reaction
(B)	$2NaH(s) \rightarrow 2Na(s) + H_2(g)$	(II)	Displacement reaction
(C)	$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$	(III)	Disproportionation reaction
(D)	$2NO_2(g) + 2NaOH(aq) \rightarrow NaNO_2(aq) + NaNO_3(aq) + H_2O(\ell)$	(IV)	Combination reaction

Choose the **correct** answer from the options given below:
(1) A-III; B-II; C-IV; D-I
(2) A-I; B-III; C-IV; D-II
(3) A-IV; B-III; C-II; D-I
(4) A-IV; B-I; C-II; D-III

47. Which of the following binary mixture form maximum boiling azeotropes?
(1) $C_2H_5OH + H_2O$
(2) $C_6H_5OH + C_6H_5NH_2$
(3) $CH_3COCH_3 + C_2H_5OH$
(4) $CS_2 + CH_3COCH_3$

48. Δ_rH° of which of the following reactions represent Δ_fH° ?
(1) $C(g) + 2H_2(g) \rightarrow CH_4(g)$
(2) $\frac{1}{8}S_8(s) + O_2(g) \rightarrow SO_2(g)$
(3) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
(4) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

49. Hemoglobin contains 0.3% of iron by mass. The number of iron atoms in 2 g of Hemoglobin is: (Given: atomic mass of Fe is 56 u)
(1) 4.56×10^{20} (2) 5.46×10^{18}
(3) 6.45×10^{19} (4) 7.29×10^{20}

50. A reaction is non-spontaneous at all temperatures when:
(1) $\Delta_rH > 0$ and $\Delta_rS > 0$
(2) $\Delta_rH < 0$ and $\Delta_rS < 0$
(3) $\Delta_rH > 0$ and $\Delta_rS < 0$
(4) $\Delta_rH < 0$ and $\Delta_rS > 0$

51. 0.1 M aqueous solution of HCl is diluted to 10 times. The pH of resulting solution will be:
(1) 1
(2) 1.5
(3) 2
(4) 2.5

52. Which of the following can **not** act as an oxidising agent?
(1) ClO_4^-
(2) NO_3^-
(3) I^-
(4) $Cr_2O_7^{2-}$

53. **Incorrect** statement regarding Henry's law is: (where: K_H is Henry's law constant)
(1) different gases have different K_H values at same temperature.
(2) The value of K_H increases with increase of temperature.
(3) Higher the value of K_H at given pressure, higher the solubility of gas in liquids.
(4) Mole fraction of gas in the solution is proportional to the partial pressure of the gas over the solution.

54. The standard enthalpies of formation of $CO_2(g)$, $H_2O(\ell)$ and glucose(s) at 25°C are –400, –300 and –1300 kJ mol^{-1}, respectively. The standard enthalpy of combustion of glucose at 25°C is:
(1) –2700 kJ mol^{-1}
(2) –3100 kJ mol^{-1}
(3) –2300 kJ mol^{-1}
(4) –2900 kJ mol^{-1}

55. 5 moles each of $I_2(g)$ and $H_2(g)$ are heated in a sealed 2 L vessel. At equilibrium 6 moles of HI are present in vessel. The equilibrium constant (K_c) of the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is:
(1) 12 (2) 15
(3) 9 (4) 6

56. The percentage composition of carbon by mole in ethane is:
(1) 75%
(2) 40%
(3) 15%
(4) 25%

57. The entropy of a sample of a certain substance increases by 0.535 J K^{-1} on adding reversibly 0.214 J of Heat at constant temperature. The temperature of the sample is:
(1) 0.16 K (2) 0.32 K
(3) 0.4 K (4) 1.2 K

58. How many species among the following can be classified as Lewis acid?
OH^-, F^-, H^+, BCl_3
(1) 1 (2) 2
(3) 3 (4) 4

59. For the redox reaction:
$$MnO_4^-(aq) + I^-(aq) + H_2O(\ell) \rightarrow MnO_2(s) + I_2(s) + OH^-(aq)$$
The **correct** coefficients of the reactants for the balanced reaction are:

	MnO_4^-	I^-	H_2O
(1)	2	3	2
(2)	2	6	4
(3)	5	2	4
(4)	5	3	2

60. Equimolal solutions of urea and sucrose in water at 25°C have:
(1) same boiling and same freezing points
(2) different boiling and different freezing points
(3) same boiling point but different freezing point
(4) same freezing point but different boiling point

61. Which among the following is **not** a state variable?
(1) Internal energy (2) Volume
(3) Heat (4) Enthalpy

62. Given below are two statements:
Statement I: The value of equilibrium constant depends on initial concentration of reactant.
Statement II: If K_c is very large, reactants predominate over products.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

63. 8 g of NaOH is present in 40 g of solution. The molality of the solution is:
(Given: atomic mass of Na = 23 u)
(1) 4.5 m
(2) 5.75 m
(3) 6.25 m
(4) 7.15 m

64. For which of the following processes, ΔS is positive?
(1) $2H(g) \rightarrow H_2(g)$
(2) $N_2(g, 273\,K) \rightarrow N_2(g, 300\,K)$
(3) $O_2(g, 1\,atm) \rightarrow O_2(g, 5\,atm)$
(4) $H_2O(\ell, 25°C) \rightarrow H_2O(s, 0°C)$

65. The pK_a of weak acid (HA) and pK_b of weak base (BOH) are 4.5 and 4.52 respectively. The pH of an aqueous solution of corresponding Salt, BA will be:
(1) 7.02 (2) 8.12
(3) 6.99 (4) 6.19

66. Amongst the following identify the species with an atom in +6 oxidation state.
(1) $BaSO_4$
(2) $HClO_3$
(3) N_2O_5
(4) SF_4

67. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Vapour pressure of 0.1 M aqueous solution of NaCl is less than vapour pressure of pure water at 25°C.
Reason R: Vapour pressure is not a colligative property.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

68. Lattice enthalpy and enthalpy of hydration for NaCl are 788 kJ mol^{-1} and –784 kJ mol^{-1}, respectively. The enthalpy of solution of NaCl will be:
(1) 4 kJ mol^{-1}
(2) 1572 kJ mol^{-1}
(3) –4 kJ mol^{-1}
(4) –1572 kJ mol^{-1}

69. Consider the reversible reactions with their equilibrium constants:

$2A(g) \rightleftharpoons 2B(g) + C(g),\ K_1$

$\frac{A}{2}(g) \rightleftharpoons \frac{B}{2}(g) + \frac{C}{4}(g),\ K_2$

The relation between K_1 and K_2 is:

(1) $K_1 = 4K_2$

(2) $K_1 = K_2^2$

(3) $K_1 = K_2^4$

(4) $K_1 = 2K_2^4$

70. Match **List-I** with **List-II**.

List-I		List-II	
(A)	8 g of $CH_4(g)$	(I)	Occupies 22.4 L at STP
(B)	2 g $H_2(g)$	(II)	Weighs 14 g
(C)	0.5 mole of $N_2(g)$	(III)	Weighs 6.4 g
(D)	0.1 g molecules of $SO_2(g)$	(IV)	3.011×10^{24} electrons

Choose the **correct** answer from the options given below:

(1) A-III; B-II; C-IV; D-I
(2) A-IV; B-I; C-II; D-III
(3) A-IV; B-III; C-II; D-I
(4) A-I; B-III; C-II; D-IV

71. One mole of an ideal gas is expanded isothermally and reversibly from 2 L to 10 L at 25°C. The enthalpy change (in kJ) for the process is:

(1) 11.2
(2) 0
(3) 2.4
(4) 6.8

72. The solubility of $Cd(OH)_2$ in water is:
(Given: The solubility product of $Cd(OH)_2$ in water is 2.5×10^{-14})

(1) 2.25×10^{-6}
(2) 1.84×10^{-5}
(3) 3.65×10^{-7}
(4) 6.11×10^{-4}

73. One mole of C_2H_6 loses ten moles of electrons to form a new compound Y. Assuming all the carbon appears in the new compound, what is the oxidation state of carbon in Y?
(Given: There is no change in oxidation state of hydrogen).

(1) +3 (2) +2
(3) +4 (4) +1

74. An aqueous solution of salt MX_2 at a certain temperature has van't Hoff factor 2.5. The degree of dissociation for the salt is:

(1) 0.45
(2) 0.35
(3) 0.55
(4) 0.75

75. $\Delta_{sub}H^o(Li), \Delta_i H^o(Li), \Delta_{BE}H^o(F_2),\ \Delta_{lattice}H^o(LiF)$ and $\Delta_f H^o(LiF)$ are 161, 520, 154, –1047 and –617 kJ mol^{-1} respectively. The $\Delta_{eg}H^o(F)$ will be:

(1) –164 kJ mol^{-1}
(2) –212 kJ mol^{-1}
(3) –282 kJ mol^{-1}
(4) –328 kJ mol^{-1}

76. In the following equilibrium:
$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$, on adding helium gas at constant temperature and pressure, the concentration of

(1) SO_2Cl_2 will increase
(2) Cl_2 will decrease
(3) SO_2Cl_2, SO_2 and Cl_2 remain constant
(4) SO_2 will increase

77. 9 g of Al react completely with how many moles of oxygen? (Given: atomic mass of Al is 27 u)

(1) 0.50 (2) 0.75
(3) 1.25 (4) 0.25

78. 500 J of energy is transferred as heat to 0.2 mol of helium gas at 298 K and 1 atm. The increase in temperature of the gas is:
(Given: R = 8.314 J mol^{-1} K^{-1})

(1) 98.26 K
(2) 134.11 K
(3) 112.12 K
(4) 120.28 K

79. An acidic buffer is obtained on mixing:

(1) 1 L of 0.1 M HCN and 1 L of 0.1 M KOH
(2) 1 L of 0.1 M NaOH and 1 L of 0.1 M NaCl
(3) 1 L of 0.1 M HCN and 2 L of 0.1 M KOH
(4) 1 L of 0.1 M HCl and 2 L of 0.1 M NaCN

80. The difference in oxidation state of two types of bromine atoms in Br_3O_8 is:

(1) 2 (2) 3
(3) 4 (4) 5

81. The osmotic pressure associated with the fluid inside the blood cell is equivalent to that of:
(1) 0.8% (mass/volume) NaCl solution
(2) 0.7% (mass/volume) NaCl solution
(3) 0.9% (mass/volume) NaCl solution
(4) 0.6% (mass/volume) NaCl solution

82. If enthalpy of formation of $H_2(g)$ from its atoms is –436 kJ mol^{-1} and that of $N_2(g)$ is –712 kJ mol^{-1}, then average bond enthalpy of N–H bond in $NH_3(g)$ is: (Given: $\Delta_f H(NH_3) = -46\ \text{kJ mol}^{-1}$)
(1) +1102 kJ mol^{-1}
(2) +964 kJ mol^{-1}
(3) +352 kJ mol^{-1}
(4) +523 kJ mol^{-1}

83. The ratio $\left(\frac{K_p}{K_c}\right)$ for the reaction

$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$, is:
(1) $\sqrt{RT}$
(2) $\frac{1}{\sqrt{RT}}$
(3) RT
(4) $\frac{1}{RT}$

84. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Molarity of a solution changes with temperature.
Reason R: Volume of solution changes with temperature.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is not the correct explanation of A.

85. In pure H_2O solution, the ratio of dissociated water to that of undissociated water at 25°C is:
(1) 2.5×10^{-8}
(2) 1.5×10^{-10}
(3) 1.7×10^{-7}
(4) 1.8×10^{-9}

86. Minimum oxidation state of iodine is in:
(1) ICl
(2) HIO_3
(3) HI
(4) I_2

87. Given below are two statements:
Statement I: Layer test can be used to identify Br^- and I^- in laboratory.
Statement II: Iodine gives an intense blue colour with starch.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

88. A vessel at T K contains CO_2 with pressure of 0.4 atm. Some of CO_2 is converted into CO on addition of graphite. The K_p of reaction
$CO_2(g) + C(s) \rightleftharpoons 2CO(g)$, will be:
(Given: total pressure at equilibrium is 0.6 atm)
(1) 0.8 atm
(2) 0.6 atm
(3) 0.2 atm
(4) 0.4 atm

89. In a process 10 kJ of work is done on the system and 3 kJ of heat is lost by the system to the surroundings. The change in internal energy (ΔU) of the system is:
(1) 13 kJ
(2) –13 kJ
(3) –7 kJ
(4) 7 kJ

90. The widely used colligative property to determine molar masses of proteins, polymers and other macromolecules is:
(1) osmotic pressure
(2) elevation in boiling point
(3) depression in freezing point
(4) lowering of vapour pressure

91. Consider the following statements about taxonomical hierarchy and choose the option with all **correct** statements.

A. Each rank represents a unit of classification.

B. As we go higher from species to kingdom, the number of common characteristics goes on increasing.

C. Order being a lower category, is the assemblage of families which exhibit many similar characters.

D. The lowest category in the taxonomic hierarchy is species.

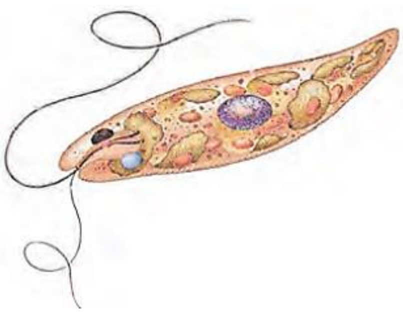
E. Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(1) A, B, C only
(2) B, C, D only
(3) A, D, E only
(4) C, D, E only

92. A cell from a plant has 2n=20 chromosomes. It undergoes meiosis. In which of the following stages of cell cycle the number of chromosomes will be same?

(1) G_1, S and G_2 phases
(2) S phase, prophase I and telophase II
(3) G_1 phase, metaphase II and anaphase II
(4) G_2 phase, interkinesis and prophase II

93. The figure below shows a type of microorganism. Select the correct statement about it.



(1) It belongs to kingdom Protista and has a cell wall made of stiff cellulose plates.
(2) It is a dinoflagellate that appears red and causes red tides.
(3) It is an obligate saprophyte.
(4) A protein-rich layer makes its body flexible.

94. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The earliest classifications of organisms were based on their uses.

Reason (R): Detailed systems of identification, nomenclature and classification were brought in by systematic and monumental description of life forms.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

95. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Resting phase	(I)	Cytokinesis
(B)	Phase of centrosome duplication	(II)	Interphase
(C)	Phase in which mitochondria are distributed to daughter cells	(III)	Interkinesis
(D)	Stage present in meiosis but not in mitosis	(IV)	S phase

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-IV, D-III
(2) A-III, B-IV, C-II, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-IV, C-I, D-III

96. Given below are two statements:

Statement I: Ascocarps and basidiocarps are fruiting bodies in which reduction division occurs, leading to formation of haploid spores.

Statement II: When a fungus reproduces sexually, two diploid hyphae of compatible mating types come together and fuse.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

97. Which of the following statements is true about the cell wall of different organisms?

A. The primary wall in plants is formed on the inner side of the cell membrane.

B. The middle lamella mainly holds or glues different neighbouring bacterial cells together.

C. In diatoms, the cell walls are embedded with chitin, making them indestructible.

D. The cell wall of fungi is primarily composed of cellulose, galactans and mannans.

E. Cell wall gives shape to the cell and protects it from mechanical damage and infection.

Choose the *most appropriate* answer from the options given below:

(1) A
(2) E
(3) B
(4) C

98. While observing a dividing cell, a scientist was not able to view Golgi complex and ER. However, after some time, both these organelles could be visualized by him. At this moment the cell is said to be present in which phase of cell cycle?

(1) Prophase
(2) Metaphase
(3) Anaphase
(4) Telophase

99. Choose the **correct** statement.

(1) Mountains and humans both are considered to be living organisms because both of them grow.
(2) Mules and male honey bees both are considered to be non-living things because both of them cannot reproduce.
(3) Unicellular *Amoeba* and multicellular hydra both are considered to be living organisms because metabolism occurs in both of them.
(4) Fungi and infertile human couples both are considered to be non-living things because both of them are not aware of themselves.

100. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The hyphae in members of class phycomycetes are said to be coenocytic.

Reason (R): In members of class phycomycetes hyphae are continuous tubes filled with multinucleated cytoplasm.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

101. The structural and functional unit of life is the cell. Which statement best explains this?

(1) All organisms are composed of cells or their products.
(2) New cells arise from pre-existing cells.
(3) Anything less than a complete structure of a cell does not ensure independent existence.
(4) The cytoplasm is the main arena of cellular activities.

102. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Order Carnivora	(I)	Cat and leopard
(B)	Family Solanaceae	(II)	Potato and species *nigrum*
(C)	Genus *Solanum*	(III)	Dog and lion
(D)	Family Felidae	(IV)	Brinjal and *Petunia*

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-IV, D-III
(2) A-III, B-IV, C-II, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-IV, C-I, D-III

103. Identify the correct statement(s) regarding the G_0 (quiescent) stage of the cell cycle.

A. Almost all human cells are present in it.

B. Cells in this stage can never return to divisional state.

C. Meristematic plant cells are mostly found in this stage.

D. The cells in this stage replicate their DNA but do not grow.

E. The cells enter this stage after exiting G_1.

Choose the *most appropriate* answer from the options given below:

(1) A, B and C
(2) E only
(3) B only
(4) A, C and E

104. Choose the **correctly** matched pair.
(1) Mesosome : ATP synthesis in bacteria
(2) Kinetochore : Formed due to secondary constriction in chromosome
(3) Leucoplasts : Single membrane bound plastids
(4) Plasmid : Genomic DNA in bacteria

105. Which of the following statements about diatoms are **correct**?
A. They are found only in marine environments.
B. Most of them are photosynthetic.
C. Their cell walls form two thin overlapping shells that fit together.
D. They are macroscopic and the chief 'producers' in the oceans.
E. Accumulation of their cell walls is referred to as 'diatomaceous earth'.
Choose the **correct** option:
(1) A, B and C
(2) B, C and D
(3) A, D and E
(4) B, C and E

106. Given below are two statements:
Statement I: The plasma membrane contains lipids, carbohydrates, proteins and nucleic acids.
Statement II: The enzymes that can digest lipids, carbohydrates, proteins and nucleic acids are present in lysosomal vesicles.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

107. Three new plants were discovered in a forest. They had morphological similarities but could not interbreed to produce viable offsprings. Which of the following taxonomical categories is least likely to be same for them?
(1) Kingdom
(2) Division
(3) Species
(4) Class

108. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Like cytoplasm of the eukaryotic cell, mitochondria can also synthesise proteins.
Reason (R): The mitochondria have ribosomes similar to those present in eukaryotic cell cytoplasm.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

109. Choose the **correct** statement.
(1) Exchange of genetic material during gamete formation leads to variations in progeny.
(2) Crossing over does not require enzymatic catalysis.
(3) Crossing over occurs in second stage of meiotic prophase I.
(4) Exchange of genetic material is immediately followed by formation of bivalents.

110. The primary drawback of Linnaeus's two-kingdom classification was that it did not distinguish between:
(1) plants and animals.
(2) unicellular and multicellular organisms.
(3) organisms with and without red blood.
(4) motile and non-motile organisms.

111. Given below are two statements:
Statement I: In anaphase I, each pole receives half the chromosome number of the parent cell.
Statement II: During anaphase I the sister chromatids separate.
In light of the above statements, choose the **correct** answer from the options given below.
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

112. Match **List-I** with **List-II**.

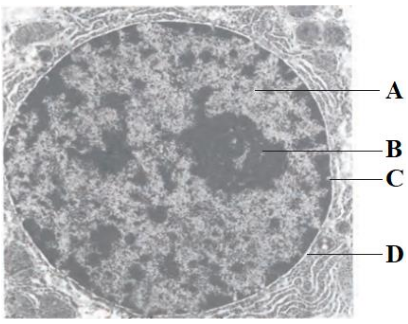
	List-I (Disease)		**List-II (Characteristic of the pathogen)**
(A)	Tobacco mosaic	(I)	Non-cellular organism
(B)	Citrus canker	(II)	Prokaryotic unicellular organism
(C)	Malaria	(III)	Eukaryotic filamentous organism
(D)	Wheat rust	(IV)	Eukaryotic unicellular organism

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-II, C-IV, D-III
(2) A-III, B-IV, C-II, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-IV, C-I, D-III

113. Read the following statements.
A. Biological names are generally in Latin and printed in italics.
B. Ernst Mayr has been called 'The Darwin of the 19^{th} century'.
C. Protozoans are considered to be primitive relatives of animals.
D. Prions possess RNA of low molecular weight.
E. All members of kingdom Plantae are partially heterotrophic.

Choose the **correct** option:
(1) A and B are incorrect
(2) A, B and C are correct
(3) C, D and E are correct
(4) B, D and E are incorrect

114. Choose the **correct** statement about the given diagram.



(1) Larger and more numerous 'C' are found in cells actively carrying out protein synthesis.
(2) 'A' is a single-membrane bound structure responsible for synthesis of rRNA.
(3) The diameter of 'B' is 10 to 50 μm.
(4) 'D' may bear 80S ribosomes on its outer side and usually remains continuous with the endoplasmic reticulum.

115. Choose the **correctly** matched pair.
(1) Plant cell wall : Inextensible
(2) Animal cell membrane : Non-permeable
(3) Cytokinesis in plant cells : By cell plate growth from periphery towards centre
(4) Cytokinesis in animal cells : By cell furrow that grows from centre towards periphery

116. The infectious agent that causes potato spindle tuber disease;
(1) is larger than a virus.
(2) consists only of free RNA.
(3) possesses capsid made of capsomeres.
(4) has cellular level of organisation.

117. Which of the following options consists of all the structures that are **not** present in bacteria?
(1) 60S ribosome, 30S ribosome and centrioles
(2) 40S ribosomes, 50S ribosomes and aleuroplasts
(3) DNA, RNA and 70S ribosomes
(4) Amyloplasts, centrosomes and cytoskeleton

118. Which of the following diseases are not caused by viruses?
A. Herpes
B. White spots on mustard
C. Creutzfeldt–Jacob disease
D. Mumps
E. Sleeping sickness

Choose the **correct** option:
(1) A and C only
(2) B and D only
(3) B, C and E only
(4) A, D and E only

119. Choose the **correct** statement.
(1) The basal body of a bacterial flagellum is a centriole-like structure.
(2) The axoneme of a eukaryotic flagellum has two doublets of centrally located microtubules.
(3) The central tubules in a eukaryotic flagellum are surrounded by central sheath.
(4) The prokaryotic and eukaryotic flagella are exactly similar in structure.

120. Which of the following pairs of taxa are **not** at the same hierarchical level?
(1) Primata and Diptera
(2) Solanaceae and Anacardiaceae
(3) Mammalia and Insecta
(4) Poales and Angiospermae

121. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The doubling of chromosome number occurs in S phase of cell cycle.

Reason (R): DNA replication occurs in the nucleus in S phase of cell cycle.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

122. Select the option that contains only components of the endomembrane system.

(1) Endoplasmic reticulum, Golgi complex, Lysosomes, Vacuoles

(2) Mitochondria, Endoplasmic reticulum, Golgi complex, Lysosomes

(3) Chloroplast, Vacuoles, Peroxisomes, Golgi complex

(4) Endoplasmic reticulum, Mitochondria, Vacuoles, Peroxisomes

123. The number of centromeres and chromatids in a pair of synapsed homologous chromosomes, respectively, is;

(1) 4 and 2.

(2) 2 and 4.

(3) 2 and 2.

(4) 4 and 4.

124. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Bacterial cell envelope	(I)	Single membrane tonoplast
(B)	Plasma membrane	(II)	No membrane
(C)	Vacuole	(III)	Three-layered structure
(D)	Centrosome	(IV)	Phospholipid bilayer

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-IV, D-III

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-IV, C-I, D-III

125. Choose the **incorrect** statement.

(1) Body organization in the members of kingdom Animalia is of tissue/organ/organ system level.

(2) Mode of nutrition in members of kingdom Fungi is obligatory heterotrophic.

(3) In the three domain system of classification, kingdom Protista was divided into two domains.

(4) The three domain system of classification has six kingdoms.

126. Arrange the following events of meiotic cell cycle in **correct** order.

A. Formation of tetrad of chromosomes.

B. Formation of tetrad of cells.

C. Formation of dyad of cells.

D. Synthesis of copy of each chromosome.

E. Synthesis of proteins required for DNA replication.

Choose the **correct** option:

(1) D – E – B – A – C

(2) E – D – A – C – B

(3) B – C – D – A – E

(4) C – A – E – B – D

127. Match **List-I** with **List-II**.

List-I		List-II	
(A)	SER	(I)	Membrane bound minute vesicles that contain various enzymes
(B)	RER	(II)	Found in cyanobacteria
(C)	Microbodies	(III)	Synthesis of lipid
(D)	Chromatophores	(IV)	Continuous with the outer membrane of the nucleus

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-IV, D-III

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-IV, C-I, D-III

128. Choose the **incorrect** statement.

(1) Carolus Linnaeus held the title '*Alexander Agassiz Professor of Zoology Emeritus*'.

(2) Carolus Linnaeus classified all living organisms into two kingdoms – Plantae and Animalia.

(3) Binomial nomenclature system was given by Carolus Linnaeus.

(4) Carolus Linnaeus was the first person to describe species '*indica*' in genus '*Mangifera*'.

129. Identify the **true** statements for fungi.

A. They show a great diversity in morphology and habitat.

B. With the exception of yeasts, fungi are filamentous.

C. The network of hyphae is known as mycelium.

D. They are cosmopolitan and occur in air, water, soil and on animals and plants.

E. The coenocytic hyphae are multinucleated.

Choose the **correct** option:

(1) A, B, C, D, E
(2) A, C, D and E only
(3) B, C and D only
(4) A, B and E only

130. Choose the **correct** pair.

(1) Mitotic apparatus : Spindle fibres + asters
(2) Bacterial cell envelope : Middle lamella + mesosome
(3) Cytoskeleton : Macrofilaments + Microtubules + Intermediate filaments
(4) Polysome : Many mRNA + one ribosome

131. Choose the **correct** statement.

(1) Most fungi reproduce sexually by sporangiospores.
(2) The sexual spores of mushroom are produced endogenously.
(3) The asexual spores of sac fungi are produced endogenously.
(4) The asexual spores in imperfect fungi are conidia.

132. Which of the following is correct regarding cell theory?

A. Schleiden and Schwann together formulated the cell theory.

B. The theory initially did not explain how new cells were formed.

C. Rudolf Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

D. One of its postulates is '*Omnis cellula-e cellula*'.

E. It states that all living organisms are composed of cells, but not their products.

Choose the **correct** option with all true statements:

(1) A, B, E only (2) A, B, C, D only
(3) B, C, D, E only (4) A, B, D, E only

133. Choose the **incorrect** statement about lichens.

(1) They have not been classified in any kingdom by Whittaker.
(2) They do not grow in polluted areas.
(3) They are symbiotic associations of two different organisms.
(4) The fungal partner is phycobiont and algal partner is mycobiont.

134. The "*cis*" and "*trans*" faces of golgi cisternae, respectively, are;

(1) forming face and the maturing face.
(2) maturing face and the forming face.
(3) concave and convex face.
(4) convex and convex face.

135. Systematics;

(1) considers only morphological features for classification of organisms.
(2) is derived from a Greek word.
(3) includes identification, nomenclature, classification, and evolutionary relationships.
(4) refers to arrangement of organisms based only on their habitat.

■■■

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Different kinds of tissues are present in an animal like human beings.
Reason R: Genetic constitution of cells from different types of tissues in human beings is always different.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

137. Oxygen (O_2) is used by organisms primarily to:
(1) break down nutrients directly through chemical reactions.
(2) assist in the indirect breakdown of nutrient molecules during respiration.
(3) extract nutrients from digested food.
(4) oxidise organic compounds directly to release energy.

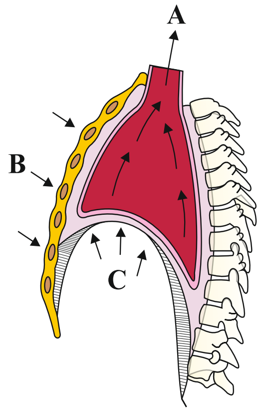
138. Carefully read the following statements and identify how many of the following statements are **correct**:
A. The electrical activity of the heart can be recorded from the body surface using a machine called electrocardiogram.
B. A heart attack and cardiac arrest are the same, as both may involve the heart stopping.
C. The oxygenated blood entering the aorta is carried by a network of arteries, venules to the tissues.
D. The flow of deoxygenated blood from the digestive organs to the liver before returning to the pulmonary circulation directly is called hepatic portal circulation.
E. Parasympathetic neural signals (a component of ANS) decrease the rate of heart beat, speed of conduction of action potential and thereby cardiac output.
(1) Only three statements are correct.
(2) Only four statements are correct.
(3) Only two statements are correct.
(4) Only one statement is correct.

139. Read the following statements.
A. *Periplaneta americana* and *Blatta orientalis* are two common Indian cockroaches.
B. Cockroaches are monoecious.
C. Kidneys of frogs are compact, dark red and bean like structures situated a little posteriorly in the body cavity on both sides of the vertebral column.
D. *Rana tigrina* is unisexual.

Identify whether the above given statements are T/F and select the **correct** option.

	A	B	C	D
(1)	F	F	F	F
(2)	T	T	F	F
(3)	F	F	F	T
(4)	T	F	T	T

140. In the given below diagram of mechanism of breathing, what does **A**, **B** and **C** depict?



(1) A-Air goes inside to lungs, B-Ribs and sternum returned to original position, C- Diaphragm contracted
(2) A-Air expelled from lungs, B-Ribs and sternum returned to original position, C-Diaphragm relaxed and arched upward
(3) A-Air expelled from lungs, B-Ribs and sternum go upward, C-Diaphragm relaxed and arched upward
(4) A-Air goes inside to lungs, B-Ribs and sternum go upward, C-Diaphragm relaxed and arched upward

141. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Superior vena cava	(I)	Carries oxygenated blood to left atrium
(B)	Inferior vena cava	(II)	Carries deoxygenated blood from lower body to right atrium
(C)	Pulmonary artery	(III)	Transports deoxygenated blood to the lungs
(D)	Pulmonary vein	(IV)	Brings deoxygenated blood from upper part of body to right atrium

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-IV, B-II, C-III, D-I
(4) A-I, B-IV, C-III, D-II

142. Read the following statements of cockroaches and choose the **correct** option.

A. Cockroaches are included in the class Arthropoda of phylum Insecta.

B. *Periplaneta americana* are about 34-53 μm long with wings that extend beyond the tip of the abdomen in males.

C. The body of cockroaches is segmented and, in each segment, the exoskeleton has hardened plates called sternites dorsally.

D. The anterior end of the head bears appendages forming biting and chewing type of mouth parts.

E. The mouth parts consist of a labium (upper lip), a pair of mandibles, a pair of maxillae and a labrum (lower lip).

(1) Only three statements are incorrect.
(2) All five statements are correct.
(3) Only four statements are incorrect.
(4) Only one statement is incorrect.

143. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Even after oxygen is delivered to body tissues, a significant amount of it remains unused in the blood.

Reason R: The unused oxygen acts as a reserve, which can be utilised during conditions like heavy exercise when oxygen demand increases.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

144. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The four-chambered heart of birds is superior to the three-chambered heart of reptiles.

Reason R: Complete double type of circulatory system ensures complete separation of oxygenated and deoxygenated blood.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

145. Division of labour is **not** exhibited by:

(1) *Paramoecium.*
(2) *Hydra.*
(3) *Taenia.*
(4) *Periplaneta.*

146. Blood carries the CO_2 in three forms. The **correct** percentages of CO_2 in these forms are:

	As carbamino haemoglobin in RBCs	**As bicarbonates**	**Dissolved form in plasma**
(1)	20-25%	70%	7%
(2)	70%	20-25%	7%
(3)	20-25%	7%	70%
(4)	7%	20-25%	70%

147. What is the functional significance of the delay of electrical impulses at the atrioventricular (AV) node during the cardiac cycle?

(1) It prevents the atria from contracting.
(2) It allows the ventricles to contract before the atria.
(3) It ensures that the atria completely empty their blood into the ventricles before ventricular contraction.
(4) It speeds up the transmission of impulses to the Purkinje fibers.

148. Which of the following is **not** a correct difference between bone and cartilage?

	Character	**Bone**	**Cartilage**
(1)	Cells	Osteocytes	Chondrocytes
(2)	Lacunae	Interconnected by minute channels called canaliculi	Canaliculi are absent
(3)	Haemopoiesis	Occurs in some bone	Does not occur in any cartilage
(4)	Matrix	Solid and pliable	Hard and non-pliable

149. Why is carbon monoxide exposure dangerous for animals?

(1) It hinders the movement of carbon dioxide in the body.
(2) It impedes the delivery of oxygen throughout the body.
(3) It enhances the transport of carbon dioxide.
(4) It breaks down haemoglobin molecules.

150. Observe the diagram below showing blood transfusion between a donor and a recipient:

Donor (Blood Group B)
↓ Transfusion
Recipient (Blood Group A)

Based on the diagram, what is the most likely outcome of this transfusion?
(1) Safe transfusion; no agglutination occurs.
(2) Agglutination occurs due to the recipient's anti-A antibodies.
(3) Agglutination occurs due to the recipient's anti-B antibodies attacking donor RBCs.
(4) No immune response because both blood groups are compatible.

151. Given below are two statements:
Statement I: Squamous epithelium is called tessellated epithelium because its flat, tile-like cells fit tightly together, forming a mosaic or tiled appearance under the microscope.
Statement II: Squamous epithelium is avascular in nature and forms a blood air barrier in the lungs.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

152. Match **List-I** with **List-II.**

	List-I		List-II
(A)	Inspiratory Capacity	(I)	Total air, a person can inspire after normal expiration.
(B)	Oxygen	(II)	Cannot measure residual volume.
(C)	Functional Residual Capacity	(III)	Volume of the air that will remain in lungs after a normal expiration.
(D)	Spirometer	(IV)	Has no significant role in regulation of respiration.

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-I, B-IV, C-III, D-II
(3) A-I, B-IV, C-II, D-III
(4) A-IV, B-I, C-III, D-II

153. After exercising regularly for several months, the resting heart rate decreases, but the cardiac output at rest remains unchanged. The heart, like any other muscle, becomes stronger through regular exercise. A stronger heart would have a greater stroke volume, allowing it to pump the same amount of blood with fewer beats.
Which of the following **best** describes the validity of the statements?
(1) The statement is true regarding cardiac output but false regarding stroke volume.
(2) The statement is true regarding both cardiac output and stroke volume.
(3) The statement is false regarding cardiac output but true regarding stroke volume.
(4) The statement is false regarding both cardiac output and stroke volume.

154. When there is no air in initial bronchioles, they do not collapse. It is due to presence of;
(1) lecithin.
(2) incomplete cartilaginous rings.
(3) complete cartilaginous rings.
(4) transformed elastin cartilage.

155. Select the **correct** sequence for O_2 movement in lungs from alveolar air to blood.
A. Capillary basement membrane
B. Capillary endothelial cells
C. Interstitial fluid
D. Alveolar basement membrane
E. Alveolar epithelial cells
(1) C → A → B → D → E
(2) A → D → B → C → E
(3) E → D → C → A → B
(4) B → D → C → E → A

156. Given below are two statements:
Statement I: The complication known as *Erythroblastosis foetalis* always arises when a Rh^{+ve} foetus develops in the womb of a Rh^{-ve} mother.
Statement II: An increase in the number of white blood cells (leukocytes) may indicate that the person is combating an infection.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

157. Choose the **incorrect** match.

(1)	Brush-bordered cuboidal epithelium	PCT of nephron
(2)	Brush-bordered columnar epithelium	Intestine of alimentary canal
(3)	Keratinized compound epithelium	Wet surface of skin
(4)	Simple ciliated epithelium	Fallopian tubes and bronchioles

158. Given below are two statements:

Statement I: The air present between nostrils and terminal bronchioles is not involved in gaseous exchange.

Statement II: Because of the presence of residual volume, alveoli typically remain inflated and do not collapse.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

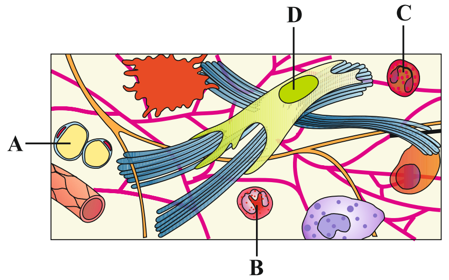
159. Match **List-I** with **List-II** w.r.t. human beings.

List I (Type of Leucocyte)		List II (Shape of Nucleus)	
(A)	Neutrophil	(I)	Kidney-shaped
(B)	Lymphocyte	(II)	Spherical or round
(C)	Monocyte	(III)	Multilobed (3-5 lobes)
(D)	Eosinophil	(IV)	Bilobed

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-II, C-IV, D-I

160. Which of the following labelled cells of loose connective tissue are signet ring shaped in structure, responsible for storage of triglycerides and are present in both areolar and adipose connective tissue?

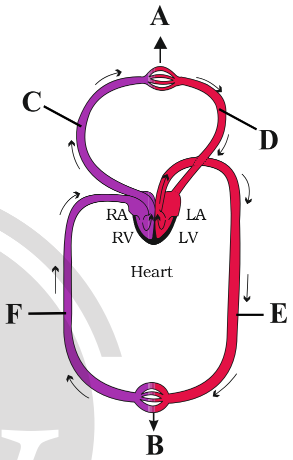


(1) A (2) B
(3) C (4) D

161. Which of the following options is **incorrect**?

(1) Partial pressure of O_2 in alveolar air is less than atmospheric air.
(2) Combining capacity of O_2 with Hb is more as compared to CO_2 at alveolar site.
(3) Oxygen dissociation curve is sigmoid shaped.
(4) Deoxygenated blood is less acidic as compared to oxygenated blood.

162. Identify A-F.



Choose the **correct** option.

(1) A–Lungs, B–Body parts, C–Pulmonary vein, D–Pulmonary artery, E–Dorsal aorta, F–Vena cava
(2) A–Lungs, B–Body parts, C–Pulmonary artery, D–Pulmonary vein, E–Dorsal aorta, F–Vena cava
(3) A–Lungs, B–Body parts, C–Pulmonary artery, D–Pulmonary vein, E–Vena cava, F–Dorsal aorta
(4) A–Body parts, B–Lungs, C–Pulmonary artery, D–Pulmonary vein, E–Vena cava, F–Dorsal aorta

163. Cockroaches are able to survive for as long as one week without a head. This is because the head holds a bit of the nervous system while the rest is situated on the dorsal side of the body.

(1) The statement is true for survival but false for part of the nervous system in the head.
(2) The statement is true for both the survival and location of the rest of nervous system.
(3) The statement is false for the location of the rest part of nervous system but true for survival and part of nervous system in head.
(4) The statement is false for both the survival and part of the nervous system in head.

164. Which of the following sets of conditions will cause the oxygen-haemoglobin dissociation curve to shift to the right, indicating a decrease in haemoglobin's affinity for oxygen?
(1) High pO_2, low pCO_2, low H^+ concentration, lower temperature
(2) High pO_2, high pCO_2, high H^+ concentration, higher temperature
(3) High pCO_2, high H^+ concentration, higher temperature
(4) Low pCO_2, low H^+ concentration, lower temperature, low pO_2

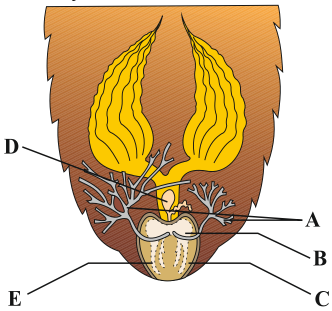
165. Which of the following statements is/are **incorrect** about the lymph?
I. Lymph is a red coloured fluid containing large proteins and formed elements.
II. Lymph is formed from tissue fluid that enters lymphatic capillaries.
III. It contains specialised lymphocytes which are responsible for the immunity of the body.
IV. Lymph is an important carrier for nutrients and hormones.
V. Fats are absorbed through the lymph in the lacteals present in the intestinal villi.
Choose the **correct** option.
(1) Only I (2) II and III
(3) III and IV (4) Only IV

166. Two cockroaches (A) and (B) are being studied. Out of them, one is male and one is female. Cockroach (A) has one pair of short thread-like structure and one pair of long and thick jointed filamentous structure at the hind end.
The other cockroach (B) has only one pair of filamentous structures arising from the 10th tergum.
Which cockroach among the given is male and female respectively?
(1) A and B
(2) Information is not sufficient
(3) B and A
(4) Cockroaches do not show sexual dimorphism

167. A person travels to a high-altitude area (above 3,500 meters) and experiences breathlessness and fatigue. Which of the following **correctly** describes the primary physiological cause of this condition?
(1) The tissues are unable to utilize the oxygen delivered to them due to an enzymatic inhibition.
(2) The total amount of haemoglobin in the blood is significantly reduced.
(3) The partial pressure of oxygen in the inspired atmospheric air is low, leading to decreased alveolar pO_2.
(4) The blood flow to the tissues is sluggish, leading to inadequate oxygen delivery despite normal arterial pO_2.

168. Which of the following **correctly** matches the ECG wave with its corresponding cardiac event?
(1) P-wave – Ventricular depolarisation
(2) U wave – Wave before T-wave
(3) T-wave – Atrial depolarisation
(4) QRS complex – Ventricular depolarisation

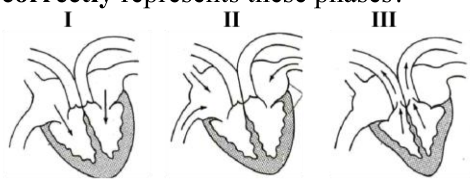
169. Identify **A** to **E** in the given diagram of the female reproductive system of cockroaches.



(1) A-Collaterial glands, B-Vestibulum, C-Genital chamber, D-Spermatheca, E-Gonapophysis
(2) A-Vestibulum, B-Collaterial gland, C-Gonapophysis, D-Spermatheca, E-Genital chamber
(3) A-Collaterial gland, B-Genital chamber, C-Vestibulum, D-Spermatheca, E-Gonapophysis
(4) A-Genital chamber, B-Spermatheca, C-Collaterial gland, D-Gonapophysis, E-Vestibulum

170. Consider the following statements.
I. Our lungs do not get overstretched to the limit of bursting due to forced inhalation due to a protective mechanism.
II. Cigarettes can cause emphysema and even cause lung cancer.
III. Neural signals from pneumotaxic centers can increase the duration of inspiration.
IV. The contraction of internal intercostal muscles lifts up the ribs and sternum.
V. Workers working in grinding and stone industries may suffer from lung fibrosis.
Which of the above statements is/are **correct**?
(1) II, III and V only
(2) II, III and IV only
(3) I, II, III and IV only
(4) I, II and V only

171. The given diagram illustrates three phases of the cardiac cycle. Which of the following sequences **correctly** represents these phases?



(1) II → III → I (2) I → II → III
(3) II → I → III (4) III → I → II

172. Match **List-I** with **List-II** w.r.t. cockroach.

List-I		List-II	
(A)	Mushroom gland	(I)	6 in number
(B)	Abdominal ganglion	(II)	9^{th} sternum
(C)	Phallomeres	(III)	6^{th} -7^{th} segment
(D)	Total abdominal segments in male	(IV)	10 segments

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-I, C-II, D-IV
(3) A-IV, B-II, C-I, D-III
(4) A-I, B-II, C-III, D-IV

173. How many of the following statements is/are **correct**?
I. The wall of bronchi and initial bronchioles are supported by dorsally incomplete cartilaginous rings.
II. When an individual move from plain to high altitude, RBCs production increases.
III. The left lung has a cardiac notch.
IV. Forceful expiration is a passive process.
(1) Three (2) Four
(3) One (4) Two

174. What should be the time interval between the "lub" and "dub" heart sounds?
(1) 0.1 sec
(2) 0.3 sec
(3) 0.5 sec
(4) 0.8 sec

175. Given below are the statements depicting functions of different parts of the alimentary canal of cockroaches. Correlate these functions with their respective organs.
I. Grinding of food particles.
II. Secretion of digestive juices.
III. Clearing of haemolymph.
The **correct** set of organs is
(1) I. Malpighian tubule II. Proventriculus III. Hepatic caeca
(2) I. Gizzard II. Gastric caeca III. Malpighian tubule
(3) I. Gastric caeca II. Gizzard III. Malpighian tubule
(4) I. Gizzard II. Crop III. Malpighian tubule

176. Blood carbon dioxide levels determine the pH of other body fluids as well as blood, including the pH of cerebrospinal fluid. How does this enable the organism to control breathing?
(1) The brain directly measures and monitors carbon dioxide and causes breathing changes accordingly.
(2) The medulla monitors pH and uses this measure to control breathing.
(3) The brain alters the pH of the cerebrospinal fluid to force the animal to retain more or less carbon dioxide.
(4) Stretch receptors in the lungs cause the medulla to speed up or slow breathing.

177. A person experiencing lung congestion is most likely suffering from:
(1) Angina pectoris.
(2) Heart attack.
(3) Cardiac arrest.
(4) Heart failure.

178. Find out the **incorrect** statement w.r.t. frog.
(1) Only tympanum can be seen externally as ear.
(2) A mature female frog can lay 2500 to 3000, unfertilised ova at a time.
(3) Bidder's canal communicates with the oviduct.
(4) Vasa efferentia are 10-12 in number.

179. Consider the following statements:
I. A healthy individual, on average, breathes around 12 to 16 times per minute.
II. The amount of air exchanged during breathing can be measured using a spirometer.
III. The diaphragm plays a significant role in both inhalation and exhalation.
Which of the above statements are **correct**?
(1) I and II only (2) I and III only
(3) II and III only (4) I, II and III

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: A molecule of carbon dioxide originating in the right thumb and exiting the body through exhalation must pass through at least two capillary beds.
Reason R: The carbon dioxide molecule first diffuses from the thumb's tissues into local capillaries, travels to the right side of the heart, and then reaches the lungs where it enters pulmonary capillaries and diffuses into alveoli to be exhaled.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

ANSWER KEY

1.	(3)
2.	(3)
3.	(4)
4.	(4)
5.	(3)
6.	(2)
7.	(4)
8.	(4)
9.	(1)
10.	(3)
11.	(3)
12.	(2)
13.	(3)
14.	(1)
15.	(4)
16.	(2)
17.	(1)
18.	(2)
19.	(4)
20.	(3)
21.	(2)
22.	(1)
23.	(1)
24.	(2)
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ANSWER KEY

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47.	(2)
48.	(2)
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ANSWER KEY

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92. (1)
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ANSWER KEY

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137. (2)
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176. (2)
177. (4)
178. (3)
179. (4)
180. (3)

HINTS & SOLUTIONS

1. (3)

$v(t) = t^2 - 3t + 2 = (t-1)(t-2)$

so, v = 0 at t = 1 s and t = 2 s

Now, $\int_0^x dx = \int_0^t v dt = \int_0^t (t^2 - 3t + 2) dt$

$x = \frac{t^3}{3} - \frac{3t^2}{2} + 2t \qquad ...(1)$

At t = 1 s, $x = \frac{1}{3} - \frac{3}{2} + 2 = \frac{5}{6}$ m

So, distance covered in between

t = 0 to t = 1 s, $\ell_1 = \frac{5}{6}$ m

At t = 2 s, $x = \frac{8}{3} - 6 + 4 = \frac{2}{3}$ m

so, distance covered in $1\text{s} \le t \le 2\text{s}$

$\ell_2 = \frac{5}{6} - \frac{2}{3} = \frac{1}{6}$ m

At t = 3 s, $x = 9 - \frac{27}{2} + 6 = \frac{3}{2}$ m

so, distance covered in $2\text{s} \le t \le 3\text{ s}$

$\ell_3 = \frac{3}{2} - \frac{2}{3} = \frac{5}{6}$ m

Total distance covered in

$0 \le t \le 3$ s

$= \ell_1 + \ell_2 + \ell_3 = \frac{5}{6} + \frac{1}{6} + \frac{5}{6} = \frac{11}{6}$ m

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2. (3)

If a particle is at rest, its acceleration may or may not be zero.

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3. (4)

Retardation: $a = \frac{dv}{dt} = -\alpha v \Rightarrow v = v_0 e^{-\alpha t}$

At $t = \frac{1}{\alpha} : v = v_0 e^{-1} = v_0 / e$

Total distance: $s = \int_0^\infty v_0 e^{-\alpha t} dt = \frac{v_0}{\alpha}$

Speed tends to zero only as $t \to \infty$

$v = ev_0$ at $t = \frac{1}{\alpha}$ is wrong

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4. (4)

$x(t) = \sqrt{1+t^2}$

$v = \frac{dx}{dt} = \frac{t}{\sqrt{1+t^2}} = \frac{t}{x}$

$a = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{t}{x}\right) = \frac{1}{x} - \frac{t}{x^2}\frac{dx}{dt}$

$= \frac{1}{x} - \frac{t}{x^2}\left(\frac{t}{x}\right) = \frac{1}{x} - \frac{t^2}{x^3} = \frac{x^2 - t^2}{x^3}$

But $x^2 - t^2 = 1$ so, $a = \frac{1}{x^3}$

Correct option (4)

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5. (3)

Systematic error is a fixed bias that affects every reading equally, so repeating the measurement N times does not change its magnitude. Hence, after 100 observations, the systematic error remains s.

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6. (2)

Side l = 2.00 ± 0.01 cm

Surface area

$A = 6l^2 = 6(2.00)^2 = 24.0\text{ cm}^2$

Relative error in A:

$\frac{\Delta A}{A} = 2\frac{\Delta l}{l} = 2\frac{0.01}{2.00} = 0.01$

$\therefore \Delta A = 24.0 \times 0.01 = 0.24\text{ cm}^2$

As there are three significant figures in length, so ΔA = 0.2

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7. (4)

Since acceleration depends only on position,

$a = \frac{3x^2}{2}$

At $t = 0 : x = 0 \Rightarrow a = 0, v = 0$

With a = 0 and v = 0 the particle never starts, so it stays at x = 0 with zero speed for all time.

$v(\text{at } t = 2\text{ s})$ = 0 m s^{-1}

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8. (4)

For projectile motion with constant downward acceleration $-g\hat{j}$,

$$y = x\tan\theta - \frac{gx^2}{2u_x^2},$$

where $u_x = u\cos\theta$ and $\tan\theta = u_y / u_x$.

Given $y = 2x - 4x^2$:

$$\tan\theta = 2,\ \frac{g}{2u_x^2} = 4 \Rightarrow u_x^2 = \frac{g}{8}.$$

Then $u_y = u_x \tan\theta = 2u_x$.

$$u = \sqrt{u_x^2 + u_y^2} = u_x\sqrt{1+\tan^2\theta}$$

$$= \sqrt{\frac{g}{8}}\sqrt{1+4} = \sqrt{\frac{5g}{8}}$$

Hence, when the particle passes through the origin, its speed is $\sqrt{\frac{5g}{8}}$.

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9. (1)

Density of the wire

$$\rho = \frac{m}{\pi r^2 l}$$

Percentage errors

$$\frac{\Delta m}{m}\times 100 = \frac{0.04}{2.00}\times 100 = 2\%$$

$$\frac{\Delta r}{r}\times 100 = \frac{0.02}{1}\times 100 = 2\%$$

$$\frac{\Delta l}{l}\times 100 = \frac{0.04}{2.00}\times 100 = 2\%$$

From error propagation

$$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 2\frac{\Delta r}{r} + \frac{\Delta l}{l} = 2\% + 2(2\%)$$

$+ 2\% = 8\%$

Maximum percentage error in density = 8%

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10. (3)

A particle starts from rest and moves with acceleration $a = 2t$ m s^{-2}.

Velocity at $t = 1$ s:

$$v = \int_0^t 2t\,dt = t^2$$

$$v(\text{at } t = 1\text{s}) = 1 \text{m s}^{-1}$$

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11. (3)

$$v_x = ky,\ v_y = kx \Rightarrow \frac{dy}{dx} = \frac{v_y}{v_x} = \frac{x}{y}$$

$$ydy = xdx \Rightarrow \frac{1}{2}y^2 = \frac{1}{2}x^2 + C$$

$$y^2 = x^2 + \text{constant}$$

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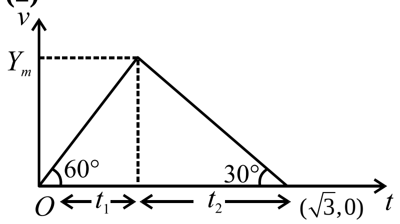
12. (2)



Let t_1 and t_2 be the time for which particle is speeding up and slowing down respectively. Then

$$\tan 60° = \frac{v_m}{t_1}$$

$$\Rightarrow t_1 = \frac{v_m}{\sqrt{3}} \qquad ...(1)$$

$$\tan 30° = \frac{v_m}{t_2}$$

$$\Rightarrow t_2 = \sqrt{3}v_m \qquad ...(2)$$

But $t_1 + t_2 = \sqrt{3}$

$$\Rightarrow \frac{v_m}{\sqrt{3}} + \sqrt{3}v_m = \sqrt{3}$$

$$\Rightarrow v_m(1+3) = 3$$

$$\Rightarrow v_m = \frac{3}{4}\text{m s}^{-1}$$

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13. (3)

For $0 \le t \le 16$

$$v(t) = |t-8| = \begin{cases} 8-t, & 0 \le t \le 8 \\ t-8, & 8 \le t \le 16 \end{cases}$$

Total displacement

$$s = \int_0^8 (8-t)dt + \int_8^{16} (t-8)dt$$

$$= \left[8t - \frac{t^2}{2}\right]_0^8 + \left[\frac{t^2}{2} - 8t\right]_8^{16} = 32 + 32$$

= 64 m

Average velocity

$$\bar{v} = \frac{s}{\Delta t} = \frac{64}{16} = 4\,\text{m s}^{-1}.$$

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14. (1)

Given $\varepsilon = a\sqrt{F} + bx^2$

From principle of homogeneity each term must be having same dimensions.

so, $[\varepsilon] = [a\sqrt{F}] = [bx^2]$

$$\Rightarrow \left[\frac{a}{b}\right] = \left[\frac{x^2}{\sqrt{F}}\right] = \left[\frac{L^2}{[MLT^{-2}]^{\frac{1}{2}}}\right]$$

$$\Rightarrow \left[\frac{a}{b}\right] = [M^{-\frac{1}{2}}L^{\frac{3}{2}}T]$$

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15. **(4)**

Statement I: The relation $\bar{v} = \frac{u+v}{2}$ holds only when acceleration is uniform. For arbitrary motion in a straight line it need not be true ⇒ Statement I is false.

Statement II: With constant acceleration the average velocity over an interval is $\frac{u+v}{2}$.

It can be zero if $v = -u$ (e.g. a body thrown up and coming back to the same point under gravity), so it can be zero ⇒ Statement II is false. Hence both statements are incorrect.

Answer: (4) Both Statement I and Statement II are incorrect.

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16. **(2)**

For minimum crossing time the swimmer points straight across the stream, so

component across river (⊥ flow):

$v_{\perp} = \sqrt{5}\,\text{m s}^{-1}$

downstream component = speed of water u

Ground-speed magnitude (constant during the trip)

$v_{avg} = \sqrt{v_{\perp}^2 + u^2} = \sqrt{5+u^2}$

Given $v_{avg} = 3$ m s^{-1}:

$\sqrt{5+u^2} = 3 \Rightarrow u^2 = 9-5 = 4$

$\Rightarrow u = 2\,\text{m s}^{-1}$

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17. **(1)**

(A) Angle between $\vec{v}_A$ and $\vec{v}_B$ is 90°. so, $|\vec{v}_{AB}| = \sqrt{v_A^2 + v_B^2}$

$\Rightarrow |\vec{v}_{AB}| = \sqrt{100+400} = \sqrt{500} = 10\sqrt{5}\,\text{m s}^{-1}$

(B) Angle between $\vec{v}_A$ and $\vec{v}_B$ is 90°. so, $|\vec{v}_{AB}| = \sqrt{v_A^2 + v_B^2}$

$\Rightarrow |\vec{v}_{AB}| = \sqrt{(20)^2 + (20)^2} = 20\sqrt{2}\,\text{m s}^{-1}$

(C) $\vec{v}_A = 10\hat{j};\ \vec{v}_B = (16\hat{i} - 12\hat{j})$

$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = -16\hat{i} + 22\hat{j}$

$|\vec{v}_{AB}| = \sqrt{(16)^2 + (22)^2} = \sqrt{256+484}$

$= \sqrt{740}\,\text{m s}^{-1}$

(D) $|\vec{v}_{AB}| = \sqrt{v_A^2 + v_B^2} = \sqrt{100+900}$

$= 10\sqrt{10}\,\text{m s}^{-1}$

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18. **(2)**

Vertical coordinates:

$y(t) = 8t - 2t^2 = -2(t^2 - 4t)$

This downward-opening parabola attains its maximum at

$t = \frac{-b}{2a} = \frac{8}{4} = 2\text{s}.$ (time of ascent)

$y_{\max} = 8(2) - 2(2)^2 = 16 - 8 = 8\text{m}$

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19. **(4)**

Assertion A: Shortest path = quickest path while crossing a river.

False. The path that gives the least time is obtained by pointing the swimmer straight across (so the whole swimming speed goes into crossing the width). That path is not the straight-line path to the point directly opposite the start.

Reason R: Because distance is minimum in the shortest path, time is minimum.

Also false. Time depends on both distance and the component of velocity along that distance; the minimum-distance route need not give the maximum crossing component of velocity, so it need not minimise time.

Answer: (4) Both A and R are false.

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20. **(3)**

For motion in a straight line

$a\,dx = v\,dv$

$\Rightarrow \int_{x=0}^{8} a\,dx = \frac{1}{2}(v^2 - v_0^2)$

Area under the a-x curve

$\int a\,dx = 2+8+2 = 12$

Initial speed $v_0 = 2\,\text{m s}^{-1}(v_0^2 = 4)$:

$\frac{1}{2}(v^2 - 4) = 12 \Rightarrow v^2 = 28 \Rightarrow v$

$= 2\sqrt{7}\,\text{m s}^{-1}$

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21. **(2)**

Treat a as a constant

$\int (e^{ax} - e^{-ax})dx = \frac{1}{a}e^{ax} - \frac{1}{-a}e^{-ax} + C$

$= \frac{e^{ax} + e^{-ax}}{a} + C$

22. (1)

From the $v_y - t$ graph the vertical component falls linearly to zero at t = 2 s.

$$u_y = gt = 10 \times 2 = 20 \text{ m s}^{-1}$$

Since t = 2 s is the time to reach the top, the total time of flight is

$$T = 2 \times 2 = 4\text{s}$$

The $v_x - t$ graph shows a constant horizontal speed

$$v_x = 8\text{ms}^{-1}$$

Horizontal range:

$$R = v_x T = 8 \times 4 = 32\text{m}$$

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23. (1)

The line joins (0, 2) and (4, 0)

Slope $m = \frac{0-2}{4-0} = -\frac{1}{2}$.

Equation: $y = mx + c$

$\Rightarrow y = -\frac{1}{2}x + 2.$

Re-arranging: $x + 2y - 4 = 0$

24. (2)

Maximum height

$$H = \frac{u^2 \sin^2\theta}{2g} \le 20\text{m } (u = 40, g = 10)$$

$$\sin^2\theta \le \frac{2gH}{u^2} = \frac{400}{1600} = 0.25 \Rightarrow \theta \le 30°$$

Range $R = \frac{u^2 \sin 2\theta}{g}$ is largest for the greatest allowed θ: $R_{\max} = \frac{40^2}{10}\sin 60° = 160\frac{\sqrt{3}}{2} = 80\sqrt{3}\text{ m}$

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25. (2)

Stone 1: $y_1 = 15t - \frac{1}{2}gt^2$ (Taking origin at cliff-edge)

Hits ground ($y_1 = -200$)

$\Rightarrow 15t - 5t^2 = -200 \Rightarrow t_1 = 8\text{ s}$

Stone 2: $y_2 = 100t - \frac{1}{2}gt^2$

At t = 8 s: separation

$= y_2 - y_1 = 85t = 85(8) = 680\text{m}$

After t_1, stone 1 stays at y = –200 m

separation $S = y_2 + 200$

Stone 2 reaches maximum height $t = 100/g = 10$ s,

$y_2^{\max} = 500\text{m}$

$s_{\max}$ = 500 + 200 = 700 m

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26. (2)

For $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$:

(a) $\vec{A} + \vec{C} = -(\vec{B} + \vec{D}) \Rightarrow$ magnitudes equal $\Rightarrow$ true

(b) Vectors need not be zero-only their sum is $\Rightarrow$ false

(c) $|\vec{A}| = |\vec{B} + \vec{C} + \vec{D}| \le |\vec{B}| + |\vec{C}| + |\vec{D}|$ (triangle inequality) $\Rightarrow$ true

(d) $\vec{B} + \vec{C} = -(\vec{A} + \vec{D})$; the sum $\vec{A} + \vec{D}$ lies in the plane defined by $\vec{A}, \vec{D}$ (or along their line if collinear), so $\vec{B} + \vec{C}$ does likewise $\Rightarrow$ true

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27. (2)

Write $y = 3\sin x + 4\cos x = 5\left(\frac{3}{5}\sin x + \frac{4}{5}\cos x\right)$

$= 5\sin(x + \theta)$

Hence y is in between ±5.

Minimum value : –5

28. (4)

In uniform circular motion only the speed (magnitude of velocity) stays constant.

Both the velocity vector (its direction keeps changing) and the acceleration vector (centripetal, continuously changing direction) are non-uniform.

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29. (4)

Uniform circular motion: centripetal acceleration

$a = -v^2/r\ \hat{r} = -v^2\ \vec{r}/r^2$

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30. (2)

As $|\vec{v}|$ is speed, so, $\Delta|\vec{v}|$ is change in speed which is zero in UCM.

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31. (3)

For 20 oscillations: $t = (40 \pm 1)$s

Time period $T = \frac{t}{20}$

Relative (percentage) error in unchanged by division by a constant:

$\frac{\Delta T}{T} \times 100\% = \frac{\Delta t}{t} \times 100\% = \frac{1}{40}$

$\times$ 100% = 2.5%

Percentage error = 2.5%

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32. **(3)**

On an $x-t$ plot an intersection of two curves means the two particles are at the same position at that instant; nothing more.

At $t = t_1$ the curves cross, so only their positions coincide (Statement 3).

Equality of slopes (velocities) is not required at a crossing, so (1 and 4) is not guaranteed true.

Their displacements from their respective starts differ, so (2) is false.

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33. **(3)**

Given $v(x)=\sqrt{81+4x}$

$$\frac{dv}{dx}=\frac{2}{\sqrt{81+4x}}$$

$$a=v(dv/dx)=\sqrt{81+4x}.\,2/\sqrt{81+4x}=2\,\text{m s}^{-2}$$

(constant)

Therefore $a\propto x^0$

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34. **(1)**

Significant figures: "5.0" has two significant digits (5 and the trailing zero after the decimal).

Order of magnitude: Write the number as

$a\times10^n$ with $1\le a\le10.$.

Here $5.0 \times 10^3 \Rightarrow a = 5.0$.

Rule says round a to 1 when $a\le5$ and to 10 when $a>5$. Since $a=5.0\le5$, , we keep the exponent $n = 3$.

Thus the order of magnitude is 3 and the number of significant figures is 2.

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35. **(1)**

For component of $\vec{A}$ along $\vec{B}$ use projection:

$$\vec{A}_{\parallel B}=\left(\frac{\vec{A}.\vec{B}}{|\vec{B}|^2}\right)\vec{B}$$

$\vec{A}=3\hat{i}+4\hat{j}+5\hat{k},\ \ \vec{B}=\hat{i}+\hat{j},$

$|B|^2=1^2+1^2=2.$

$\vec{A}.\vec{B}=3(1)+4(1)+5(0)=7$

$$\vec{A}_{\parallel B}=\frac{7}{2}(\hat{i}+\hat{j}).$$

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36. **(3)**

Because each particle always heads towards the next one of the triangle shrinks self-similarly and every particle's path spirals towards the centre.

Distance of any corner from the centre of the equilateral triangle:

$$R_0=\frac{a}{\sqrt{3}}.$$

Component of velocity towards the centre of the triangle, $=v\cos30°=\frac{\sqrt{3}}{2}v$

Time taken to reach the centre $t=\dfrac{a/\sqrt{3}}{(\sqrt{3}/2)v}=\dfrac{2a}{3v}.$

Hence distance each particle covers:

$$s=vt=v\left(\frac{2a}{3v}\right)=\frac{2a}{3}.$$

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37. **(1)**

Initial component

$u_x=u\cos60°=10\sqrt{3}(1/2)=5\sqrt{3}\ \text{m s}^{-1}$

$u_x=v_x$

$u\cos60°=v\cos30°$

$\Rightarrow v = 10\ \text{m s}^{-1}$

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38. **(2)**

Area under a-t graph gives change in velocity

change in velocity $=\frac{1}{2}\times(-10)\times2=-10$

$\Rightarrow v-(-10) = -10$

$v = -20\ \text{m s}^{-1}$

Speeds: $|v_0|=10\,\text{m s}^{-1}, |v|=20\,\text{m s}^{-1}$

Change in speed = 20 – 10

= $10\ \text{m s}^{-1}$

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39. **(4)**

In uniform circular motion

$\vec{a}$ is centripetal (towards the centre), opposite to $\vec{r}\Rightarrow\vec{a}.\vec{r}<0$

$\vec{a}$ is perpendicular to the tangential velocity $\vec{v}\Rightarrow\vec{a}.\vec{v}=0$

Thus statements (2) and (3) are correct.

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40. **(1)**

Relative displacement

$$s_{BA}=\left(20t+\frac{1}{2}at^2\right)-20t=\frac{1}{2}at^2$$

$$=\frac{1}{2}(1)(50)^2=1250\,\text{m}$$

To let the rear of train B(guard) just pass the front of train A,

if D_{initial} be the distance between two trains

$$s_{BA}=D_{\text{initial}}+2L$$

$$\Rightarrow D_{\text{initial}}=1250-800=450\,\text{m}$$

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41. **(2)**

Since

$\vec{A}+\vec{B}+\vec{C}=\vec{0}\Rightarrow\vec{C}=-(\vec{A}+\vec{B}),$

1. $\vec{A}\times\vec{C}=\vec{A}\times(-\vec{A}-\vec{B})=-\vec{A}\times\vec{B}\neq\vec{A}\times\vec{B}$

(except for trivial parallel cases) $\Rightarrow$ false.

2. $\vec{C}\times\vec{A}=(-\vec{A}-\vec{B})\times\vec{A}=-\vec{B}\times\vec{A}=\vec{A}\times\vec{B}$

(using $\vec{B}\times\vec{A}=-\vec{A}\times\vec{B}$) $\Rightarrow$ always true.

3. $\vec{B}\cdot\vec{C}=\vec{B}\cdot(-\vec{A}-\vec{B})=-\vec{A}\cdot\vec{B}-|\vec{B}|^2\neq\vec{A}\cdot\vec{B}$

in general $\Rightarrow$ false.

4. $\vec{A}\cdot\vec{C}=-|\vec{A}|^2-\vec{A}\cdot\vec{B}$ and

$\vec{B}\cdot\vec{C}=-\vec{A}\cdot\vec{B}-|\vec{B}|^2$; equal only if

$|\vec{A}|=|\vec{B}|\Rightarrow$ not generally true.

Hence only statement 2 $\vec{A}\times\vec{B}=\vec{C}\times\vec{A}$ is correct.

(NCERT Ed. 2025-26, 11th, Page No. 29)

42. **(1)**

Relative speed of the cars

$$v_{rel}=18+27=45\,\text{km h}^{-1}$$

Time until meet

$$t=\frac{\text{initial separation}}{v_{rel}}=\frac{36\text{ km}}{45\text{ km h}^{-1}}=0.8\text{ h}$$

Bird's speed v_b = 36 km h^{-1}

Distance flown by the bird

$$d=v_bt=36\times0.8=28.8\,\text{km}$$

Total distance = 28.8 km

(NCERT Ed. 2025-26, 11th, Page No. 21)

43. **(1)**

Use the cosine (parallelogram) relation for the magnitude of a difference:

$$|\vec{A}-\vec{B}|^2=|\vec{A}|^2+|\vec{B}|^2-2|\vec{A}||\vec{B}|\cos\theta,$$

where θ is the angle between $\vec{A}$ and $\vec{B}$.

Since we are told the angle between $\vec{A}$ and $-\vec{B}$ is 60°, the angle between $\vec{A}$ and $\vec{B}$ is

$$180^\circ-60^\circ=120^\circ,\quad \cos120^\circ=-\frac{1}{2}.$$

With $|\vec{A}|=|\vec{B}|=10\,\text{m}$:

$$|\vec{A}-\vec{B}|^2=10^2+10^2-2(10)(10)\left(-\frac{1}{2}\right)$$

$$=100+100+100=300\Rightarrow|\vec{A}-\vec{B}|$$

$$=10\sqrt{3}\,\text{m}.$$

(NCERT Ed. 2025-26, 11th, Page No. 34)

44. **(2)**

Inside the elevator (a non-inertial frame accelerating downward with $a_e=5\,\text{m s}^{-2}$), the ball experiences an effective downward acceleration

$$g'=g-a_e=10-5=5\,\text{m s}^{-2}$$

Dropped from rest relative to the elevator at height h = 2.5 m:

$$h=\frac{1}{2}g't^2\Rightarrow2.5=\frac{1}{2}(5)t^2$$

$$\Rightarrow t^2=1\Rightarrow t=1\,\text{s}$$

(NCERT Ed. 2025-26, 11th, Page No. 19)

45. **(4)**

For a fixed physical quantity Q,

$Q=nu\Rightarrow n$

$=\dfrac{Q}{u}$ (inverse relation)

Taking logarithms:

$$\log n=\log Q-\log u,$$

which is a straight line of slope –1 on a log n vs log u pot.

Only graph (4) shows this linear negative-slope relationship.

(NCERT Ed. 2025-26, 11th, Page No. 1)

HINTS & SOLUTIONS

46. (4)

- $3Mg(s) + N_2(g) \rightarrow Mg_3N_2(s)$: Combination reaction
- $2NaH(s) \rightarrow 2Na(s) + H_2(g)$: Decomposition reaction
- $Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(s)$: Displacement reaction
- $2NO_2(g) + 2NaOH(aq) \rightarrow NaNO_2(aq) + NaNO_3(aq) + H_2O(\ell)$: Disproportionation reaction

(NCERT Ed. 2025-26 11th Part-II Page No. 242, 244)

47. (2)

Binary mixture of C_6H_5OH and $C_6H_5NH_2$ shows negative deviation from Raoult's law. Vapour pressure of solution is less than vapour pressure of pure C_6H_5OH and $C_6H_5NH_2$, so the boiling point of solution is more than boiling point of pure phenol and aniline, hence it form maximum boiling azeotrope.

(NCERT Ed. 2025-26 12th Part-I Page No. 14)

48. (2)

$\Delta_f H^o$ is defined when one mole of a compound is formed from its elements in most stable state of aggregation.

(NCERT Ed. 2025-26 11th Part-I Page No. 149)

49. (3)

Moles of iron atom $= \dfrac{2 \times \frac{0.3}{100}}{56}$

$\therefore$ Number of iron atom $= \dfrac{2 \times 0.3}{56 \times 100} \times 6.022 \times 10^{23}$

$= 6.45 \times 10^{19}$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

50. (3)

For a non-spontaneous reaction at all temperatures,

$\Delta G > 0$

$\Delta H - T\Delta S > 0$

or $\Delta H > 0$ and $\Delta S < 0$

(NCERT Ed. 2025-26 11th Part-I Page No. 162)

51. (3)

$M_1V_1 = M_2V_2$

$0.1 \times V = M_2 \times 10\,V$

$M_2 = 0.01\,M$

$pH = -\log(0.01) = 2$

(NCERT Ed. 2025-26 11th Part-I Page No. 194)

52. (3)

I in I^- is present in its minimum oxidation state -1 so it can not be reduced further. In other words, it can not act as an oxidising agent.

(NCERT Ed. 2025-26 11th Part-II Page No. 241)

53. (3)

Higher the value of K_H at given pressure, lower the solubility of gas in liquids.

(NCERT Ed. 2025-26 12th Part-I Page No. 07)

54. (4)

$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(\ell)$

$\Delta_r H = [6 \times (-400) + 6(-300)] - [1 \times (-1300) + 0]$

$= -2900\ kJ\ mol^{-1}$

(NCERT Ed. 2025-26 11th Part-I Page No. 152)

55. (3)

	$H_2(g)$	+ $I_2(g)$	$\rightleftharpoons 2HI(g)$
(i)	5 mol	5 mol	–
(f)	$5-3=2$ mol	$5-3=2$ mol	6 mol

$$K_c = \frac{\left(\frac{6}{2}\right)^2}{\left(\frac{2}{2}\right)\left(\frac{2}{2}\right)} = 9$$

(NCERT Ed. 2025-26 11th Part-I Page No. 175)

56. (4)

C_2H_6:

Mole of carbon = 2

Mole of hydrogen = 6

% of carbon by mole $= \dfrac{2}{2+6} \times 100 = 25\%$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

57. (3)

$$T = \frac{\Delta H}{\Delta S} = \frac{0.214}{0.535} = 0.4\ K$$

(NCERT Ed. 2025-26 11th Part-I Page No. 160)

58. (2)

- Lewis base: OH^-, F^-
- Lewis acid: H^+, BCl_3

(NCERT Ed. 2025-26 11th Part-I Page No. 192)

59. (2)

$$2MnO_4^-(aq)+6I^-(aq)+4H_2O(\ell)\to 3I_2(s)+2MnO_2(s)+8OH^-(aq)$$

(NCERT Ed. 2025-26 11th Part-II Page No. 246)

60. (1)

Since ΔT_b and ΔT_f are colligative properties so equimolal solution of normal solutes will have same boiling and same freezing points.

(NCERT Ed. 2025-26 12th Part-I Page No. 16, 17)

61. (3)

State variable is a variable which is independent of the path of change of state. Heat depends on path.

(NCERT Ed. 2025-26 11th Part-I Page No. 138)

62. (4)

- The value of equilibrium constant is independent of initial concentration of the reactants and products.
- If K_c is very large, the reaction proceeds nearly to completion.
- If K_c is $< 10^{-3}$, reactants predominate over products.

(NCERT Ed. 2025-26 11th Part-I Page No. 181)

63. (3)

$$\text{Molality (m)}=\frac{\left(\frac{8}{40}\right)}{(40-8)\times10^{-3}}=6.25$$

(NCERT Ed. 2025-26 11th Part-I Page No. 24)

64. (2)

- When gaseous mole decreases, entropy decreases
  $2H(g)\to H_2(g),\quad \Delta n_g<0$
- On increasing temperature, randomness of gaseous molecules increases so entropy increases
  $N_2(g, 273\,K)\to N_2(g, 300\,K),\quad \Delta S>0$
- On increasing pressure, gaseous molecules come closer so entropy decreases
  $O_2(g, 1\,atm)\to O_2(g, 5\,atm),\quad \Delta S<0$
- On freezing, randomness of molecules decreases so entropy decreases
  $H_2O(\ell, 25°C)\to H_2O(s, 0°C),\quad \Delta S<0$

(NCERT Ed. 2025-26 11th Part-I Page No. 159)

65. (3)

For aqueous solution of salt of weak acid and weak base:

$$pH=\frac{1}{2}[pK_w+pK_a-pK_b]$$

$$=\frac{1}{2}[14+4.5-4.52]=6.99$$

(NCERT Ed. 2025-26 11th Part-I Page No. 202)

66. (1)

- $BaSO_4: +2+x+4(-2)=0$
  $x=+6$
- $HClO_3: +1+x+3(-2)=0$
  $x=+5$
- $N_2O_5: 2x+5(-2)=0$
  $x=+5$
- $SF_4: x+4(-1)=0$
  $x=+4$

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

67. (4)

Vapour pressure of 0.1 M aqueous solution of NaCl is less than vapour pressure of pure water at 25°C because solute particles cover the surface, as a result escaping tendency of water molecules decreases.

(NCERT Ed. 2025-26 12th Part-I Page No. 15)

68. (1)

$$\Delta_{sol}H^o=\Delta_{lattice}H^o+\Delta_{hyd}H^o$$

$$=788-784=4\text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 156)

69. (3)

$$2A(g)\rightleftharpoons 2B(g)+C(g),\ K_1=\frac{[B]^2[C]}{[A]^2}$$

$$\frac{A}{2}(g)\rightleftharpoons\frac{B}{2}(g)+\frac{C}{4}(g),\ K_2=\frac{[B]^{\frac{1}{2}}[C]^{\frac{1}{4}}}{[A]^{\frac{1}{2}}}$$

$$\therefore\ K_2=(K_1)^{\frac{1}{4}}$$

$$K_2^4=K_1$$

(NCERT Ed. 2025-26 11th Part-I Page No. 177)

70. (2)

- $CH_4 : \text{mole} = \frac{8}{16} = 0.5$
  No. of electrons = $0.5 \times 10 \times 6.022 \times 10^{23}$
  $= 3.011 \times 10^{24}$
- $H_2 : \text{mole} = \frac{2}{2} = 1$
  Volume at STP = 22.4 L
- $N_2 : \text{mass} = 0.5 \times 28 = 14\ g$
- $SO_2 : \text{mole} = 0.1$
  mass = $0.1 \times 64 = 6.4$ g

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

71. (2)

$\Delta T = 0$

$\therefore\ \Delta H = nC_p\Delta T = 0$

(NCERT Ed. 2025-26 11th Part-I Page No. 145)

72. (2)

$$\underset{}{Cd(OH)_2(s)} \rightleftharpoons \underset{s}{Cd^{2+}}(aq) + \underset{2s}{2OH^-}(aq)$$

$K_{sp} = (s)^1(2s)^2$

$2.5 \times 10^{-14} = 4s^3$

$25 \times 10^{-15} = 4s^3$

$s^3 = \frac{25}{4} \times 10^{-15}$

$s = 1.84 \times 10^{-5}$

(NCERT Ed. 2025-26 11th Part-I Page No. 205)

73. (2)

$C_2^{6-} \to 10e^- + 2C^x$

Apply charge balance:

$-6 = -10 + 2x$

x = +2

(NCERT Ed. 2025-26 11th Part-II Page No. 246)

74. (4)

$MX_2 \rightleftharpoons M^{2+} + 2X^-$

n = 3

$i = 1 + (n-1)\alpha$

$2.5 = 1 + (3-1)\alpha$

$1.5 = 2\alpha$

$\alpha = 0.75$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

75. (4)

$$\Delta_f H^o = \Delta_{sub}H^o + \frac{1}{2}\Delta_{BE}H^o + \Delta_i H^o + \Delta_{eg}H^o + \Delta_{lattice}H^o$$

$$-617 = 161 + \frac{154}{2} + 520 + \Delta_{eq}H^o - 1047$$

$\Delta_{eq}H^o = -328\ kJ\ mol^{-1}$

(NCERT Ed. 2025-26 11th Part-I Page No. 155)

76. (4)

On addition of inert gas at constant temperature and pressure reaction moves in that direction where number of gaseous moles increases.

(NCERT Ed. 2025-26 11th Part-I Page No. 186)

77. (4)

$2Al + \frac{3}{2}O_2 \to Al_2O_3$

mole of $Al = \frac{9}{27} = \frac{1}{3}$

$\therefore$ mole of O_2 required $= \frac{3}{4} \times \frac{1}{3} = 0.25$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

78. (4)

$q = nC_p\Delta T$

$500 = 0.2 \times \frac{5}{2}R \times \Delta T$

(For He, $C_p = \frac{5}{2}R$)

$\Delta T = \frac{1000}{R} = \frac{1000}{8.314} = 120.28\ K$

(NCERT Ed. 2025-26 11th Part-I Page No. 145)

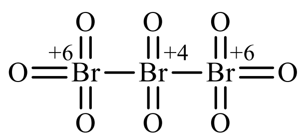
79. (4)

	HCl	+ NaCN	⟶ NaCl	+ HCN
(*i*)	0.1 mol	0.2 mol	–	–
(f)	–	0.1 mol	0.1 mol	0.1 mol

- Due to NaCl solution will be neutral.
- Due to presence of 0.1 mol of NaCN and 0.1 mol HCN solution will behave as an acidic buffer.

(NCERT Ed. 2025-26 11th Part-I Page No. 203)

80. (1)



Required difference = 6 – 4 = 2

(NCERT Ed. 2025-26 11th Part-II Page No. 245)

81. (3)
The osmotic pressure associated with the fluid inside the blood cell is equivalent to that of 0.9% (mass/volume) NaCl solution
(NCERT Ed. 2025-26 12th Part-I Page No. 22)

82. (3)
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
$\Delta_r H = \left[(BE_{N_2}) + 3(BE)_{H_2}\right] - \left[6(BE)_{N-H}\right]$
$2 \times (-46) = \left[712 + 3(436)\right] - \left[6 \times (BE)_{N-H}\right]$
$BE_{N-H} = +352 \text{ kJ mol}^{-1}$
(NCERT Ed. 2025-26 11th Part-I Page No. 154)

83. (2)
$\Delta n_g = (1) - \left(1 + \frac{1}{2}\right) = -\frac{1}{2}$
$K_p = K_c(RT)^{-\frac{1}{2}}$
$\frac{K_p}{K_c} = \frac{1}{\sqrt{RT}}$
(NCERT Ed. 2025-26 11th Part-I Page No. 178)

84. (3)
Molarity of a solution changes with temperature as volume of solution changes with temperature.
(NCERT Ed. 2025-26 11th Part-I Page No. 23)

85. (4)
Molarity of pure $H_2O = 55.55$ M
In pure water, $[H^+] = [OH^-] = 10^{-7}$ M
$\therefore$ required ratio $= \frac{10^{-7}}{55.55} = 1.8 \times 10^{-9}$
(NCERT Ed. 2025-26 11th Part-I Page No. 193)

86. (3)
- $ICl : x + (-1) = 0 \Rightarrow x = +1$
- $HIO_3 : +1 + x + 3(-2) = 0 \Rightarrow x = +5$
- $HI : +1 + x = 0 \Rightarrow x = -1$
- $I_2 : 2x = 0 \Rightarrow x = 0$

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

87. (1)
- Layer test can be used to identify Br^- and I^- in laboratory.
- Iodine gives an intense blue colour with starch.

(NCERT Ed. 2025-26 11th Part-II Page No. 249)

88. (1)

	$CO_2(g)$ + $C(s)$	$\rightleftharpoons 2CO(g)$
(*i*)	0.4 atm	–
(f)	$(0.4 - x)$ atm	2x atm

$p_{total} = 0.6 = 0.4 - x + 2x$
$0.6 = 0.4 + x$
x = 0.2 atm
$\therefore K_p = \frac{(2x)^2}{(0.4 - x)} = \frac{0.4 \times 0.4}{0.2} = 0.8 \text{ atm}$
(NCERT Ed. 2025-26 11th Part-I Page No. 180)

89. (4)
$\Delta U = q + w$
$= (-3) + (10) = 7 \text{ kJ}$
(NCERT Ed. 2025-26 11th Part-I Page No. 140)

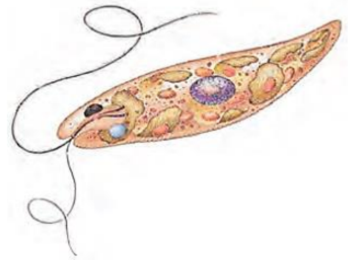
90. (1)
Measurement of osmotic pressure provides another method of determining molar masses of solutes.
(NCERT Ed. 2025-26 12th Part-I Page No. 21)

HINTS & SOLUTIONS

91. (3)
Each rank or taxon, in fact, represents a unit of classification. All organisms, including those in the plant and animal kingdoms have species as the lowest category. Genus comprises a group of related species which has more characters in common in comparison to species of other genera. As we go higher, the number of common characteristics goes on decreasing. Order being a higher category, is the assemblage of families which exhibit a few similar characters.
(NCERT Ed. 2025-26, 11th, Page No. 6)

92. (1)
S or synthesis phase marks the period during which DNA synthesis or replication takes place. During this time the amount of DNA per cell doubles. If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell had diploid or 2n number of chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 2n. Meiosis I is a reductional division. After Meiosis I, the chromosome number is halved, but each chromosome still consists of two sister chromatids. Therefore, at G_1, S and G_2 phases, the number of chromosomes will be 20. After completion of meiosis I, the haploid daughter cells are formed, in them number of chromosomes is 10. After meiosis II also the daughter cells have 10 chromosomes each.
(NCERT Ed. 2025-26, 11th, Page No. 121)

93. (4)
The diagram shows *Euglena*. Majority of the euglenoids are fresh water organisms found in stagnant water. Instead of a cell wall, they have a protein rich layer called pellicle which makes their body flexible. They have two flagella, a short and a long one. Though they are photosynthetic in the presence of sunlight, when deprived of sunlight they behave like heterotrophs by predating on other smaller organisms. Interestingly, the pigments of euglenoids are identical to those present in higher plants. Example: *Euglena*. The cell wall of dinoflagellates has stiff cellulose plates on the outer surface. Most of them have two flagella; one lies longitudinally and the other transversely in a furrow between the wall plates. Very often, red dinoflagellates (Example: *Gonyaulax*) undergo such rapid multiplication that they make the sea appear red (red tides).



Euglena
(NCERT Ed. 2025-26, 11th, Page No. 14)

94. (2)
In early days, human beings needed to find sources for their basic needs of food, clothing and shelter. Hence, the earliest classifications were based on the 'uses' of various organisms. The description of living organisms including human beings began much later in human history. Societies which indulged in anthropocentric view of biology could register limited progress in biological knowledge. Systematic and monumental description of life forms brought in, out of necessity, detailed systems of identification, nomenclature and classification.
(NCERT Ed. 2025-26, 11th, Page No. 5)

95. (4)
The interphase, though called the resting phase, is the time during which the cell is preparing for division by undergoing both cell growth and DNA replication in an orderly manner. In animal cells, during the S phase, DNA replication begins in the nucleus, and the centriole duplicates in the cytoplasm. At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells. The stage between the two meiotic divisions is called interkinesis and is generally short lived.
(NCERT Ed. 2025-26, 11th, Page No. 121)

96. (2)
When a fungus reproduces sexually, two haploid hyphae of compatible mating types come together and fuse. In some fungi the fusion of two haploid cells immediately results in diploid cells (2n). However, in other fungi (ascomycetes and basidiomycetes), an intervening dikaryotic stage (n + n, i.e., two nuclei per cell) occurs; such a condition is called a dikaryon and the phase is called dikaryophase of fungus. Later, the parental nuclei fuse and the cells become diploid. The fungi form fruiting bodies in which reduction division occurs, leading to formation of haploid spores.
(NCERT Ed. 2025-26, 11th, Page No. 16)

97. (2)

A non-living rigid structure called the cell wall forms an outer covering for the plasma membrane of fungi and plants. Cell wall not only gives shape to the cell and protects the cell from mechanical damage and infection, it also helps in cell-to-cell interaction and provides barrier to undesirable macromolecules. Algae have cell wall, made of cellulose, galactans, mannans and minerals like calcium carbonate, while in other plants it consists of cellulose, hemicellulose, pectins and proteins. The cell wall of a young plant cell, the primary wall is capable of growth, which gradually diminishes as the cell matures and the secondary wall is formed on the inner (towards membrane) side of the cell. The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together. The walls of diatoms are embedded with silica and thus the walls are indestructible. The cell walls of fungi are composed of chitin and polysaccharides.

(NCERT Ed. 2025-26, 11th, Page No. 94)

98. (4)

In the telophase, the nucleolus, golgi complex and ER reform.

(NCERT Ed. 2025-26, 11th, Page No. 124)

99. (3)

Mountains are non-living things although they grow by deposition of material on external surface. Mules and honey bee both are living organisms although mules cannot reproduce but male honey bees can. Metabolism is a defining property of living organisms. Human is the only organism who is aware of himself.

(NCERT Ed. 2022-23, 11th, Page No. 4)

100. (1)

With the exception of yeasts which are unicellular, fungi are filamentous. Their bodies consist of long, slender thread-like structures called hyphae. The network of hyphae is known as mycelium. Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called coenocytic hyphae. Others have septae or cross walls in their hyphae.

(NCERT Ed. 2025-26, 11th, Page No. 16)

101. (3)

Unicellular organisms are capable of (i) independent existence and (ii) performing the essential functions of life. Anything less than a complete structure of a cell does not ensure independent living. Hence, cell is the fundamental structural and functional unit of all living organisms.

(NCERT Ed. 2025-26, 11th, Page No. 87)

102. (2)

Dog and lion are both members of the Order Carnivora. Brinjal and *Petunia* represent two different genera within the family Solanaceae. Potato (*Solanum tuberosum*) and species *nigrum* are both species within the Genus *Solanum*. Cat (genus *Felis*) and leopard (genus *Panthera*) are both members of the Family Felidae.

(NCERT Ed. 2025-26, 11th, Page No. 7)

103. (2)

Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of plants throughout their life. S or synthesis phase marks the period during which DNA synthesis or replication takes place.

(NCERT Ed. 2025-26, 11th, Page No. 122)

104. (1)

Mesosomes help in cell wall formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content. Every chromosome (visible only in dividing cells) essentially has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present. The leucoplasts are the colourless plastids of varied shapes and sizes with stored nutrients, they are double-membrane bound. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids. The plasmid DNA confers certain unique phenotypic characters to such bacteria. One such character is resistance to antibiotics.

(NCERT Ed. 2025-26, 11th, Page No. 91)

105. (4)

Chrysophytes: This group includes diatoms and golden algae (desmids). They are found in fresh water as well as in marine environments. They are microscopic and float passively in water currents (plankton). Most of them are photosynthetic. In diatoms the cell walls form two thin overlapping shells, which fit together as in a soap box. The walls are embedded with silica and thus the walls are indestructible. Thus, diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as 'diatomaceous earth'.

(NCERT Ed. 2025-26, 11th, Page No. 14)

106. (3)

Cell membrane is mainly composed of phospholipids. Later, biochemical investigation clearly revealed that the cell membranes also possess protein and carbohydrate. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases – lipases, proteases, carbohydrases) optimally active at the acidic pH. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

(NCERT Ed. 2025-26, 11th, Page No. 96)

107. (3)

Since the three new plants have been discovered, they will all belong to kingdom Plantae. Due to their morphological similarities, they are very likely to belong to the same division and/or class. However, because they cannot interbreed, they are, by definition, different species. Therefore, this category is the least likely to be the same. Taxonomic studies consider a group of individual organisms with fundamental similarities as a species.

(NCERT Ed. 2025-26, 11th, Page No. 6)

108. (3)

The matrix of mitochondria also possesses single circular DNA molecule, a few RNA molecules, ribosomes (70S) and the components required for the synthesis of proteins. The eukaryotic ribosomes (present in eukaryotic cell cytoplasm) are 80S.

(NCERT Ed. 2025-26, 11th, Page No. 97)

109. (1)

Prophase I: Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis. It has been further subdivided into the following five phases based on chromosomal behaviour, i.e., Leptotene, Zygotene, Pachytene, Diplotene and Diakinesis. During zygotene, chromosomes start pairing together and this process of association is called synapsis. Such paired chromosomes are called homologous chromosomes. Electron micrographs of this stage, indicate that chromosome synapsis is accompanied by the formation of complex structure called synaptonemal complex. The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. However, these are more clearly visible at the next stage. The first two stages of prophase I are relatively short-lived compared to the next stage that is pachytene. During this stage, the four chromatids of each bivalent chromosomes becomes distinct and clearly appear as tetrads. This stage is characterised by the appearance of recombination nodules, the sites at which crossing over occurs between non-sister chromatids of the homologous chromosomes. Crossing over is the exchange of genetic material between two homologous chromosomes. Crossing over is also an enzyme-mediated process and the enzyme involved is called recombinase. Crossing over leads to recombination of genetic material on the two chromosomes. Meiosis increases the genetic variability in the population of organisms from one generation to the next.

(NCERT Ed. 2025-26, 11th, Page No. 128)

110. (2)

In Linnaeus' time a Two Kingdom system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms.

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111. (3)

During metaphase I the bivalents arrange on the equatorial plate. This is followed by anaphase I in which homologous chromosomes move to the opposite poles with both their chromatids. Each pole receives half the chromosome number of the parent cell. In telophase I, the nuclear membrane and nucleolus reappear. Meiosis II is similar to mitosis. During anaphase II the sister chromatids separate.

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112. (1)
Tobacco mosaic is caused by a virus (non-cellular organism). Citrus canker is caused by a bacterium (Prokaryotic unicellular organism). Malaria is caused by a protozoan (Eukaryotic unicellular organism). Wheat rust is caused by *Puccinia* (Eukaryotic filamentous organism).
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113. (4)
Ernst Mayr has been called 'The Darwin of the 20th century'. Prions consists of abnormally folded protein but lack genetic material. Kingdom Plantae includes all eukaryotic chlorophyll-containing organisms commonly called plants. A few members are partially heterotrophic such as the insectivorous plants or parasites. Bladderwort and Venus fly trap are examples of insectivorous plants and *Cuscuta* is a parasite.
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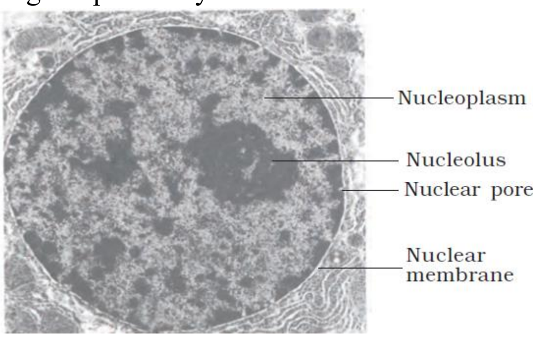
114. (4)
Electron microscopy has revealed that the nuclear envelope, which consists of two parallel membranes with a space between (10 to 50 nm) called the perinuclear space, forms a barrier between the materials present inside the nucleus and that of the cytoplasm. The outer membrane usually remains continuous with the endoplasmic reticulum and also bears ribosomes on it. At a number of places the nuclear envelope is interrupted by minute pores, which are formed by the fusion of its two membranes. These nuclear pores are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm. The nuclear matrix or the nucleoplasm contains nucleolus and chromatin. The nucleoli are spherical structures present in the nucleoplasm. The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure. It is a site for active ribosomal RNA synthesis. Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.



Structure of nucleus
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115. (1)
In an animal cell, cytokinesis is achieved by the appearance of a furrow in the plasma membrane. The furrow gradually deepens and ultimately joins in the centre dividing the cell cytoplasm into two. Plant cells are enclosed by a relatively inextensible cell wall, therefore they undergo cytokinesis by a different mechanism. In plant cells, wall formation starts in the centre of the cell and grows outward to meet the existing lateral walls. The plasma membrane is selectively permeable and facilitates transport of several molecules.
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116. (2)
In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. They are acellular organisms.
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117. (4)
The eukaryotic ribosomes are 80S while the prokaryotic ribosomes are 70S. Each ribosome has two subunits, larger and smaller subunits. The two subunits of 80S ribosomes are 60S and 40S while that of 70S ribosomes are 50S and 30S. DNA, RNA and 70S ribosomes are present in bacteria. Aleuroplasts, amyloplasts, centrosomes and cytoskeleton are not present in bacteria.
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118. (3)
Viruses cause diseases like mumps, small pox, herpes and influenza. AIDS in humans is also caused by a virus. White spots on mustard are caused by *Albugo* (a fungus). Creutzfeldt–Jacob disease is caused by a prion. Sleeping sickness is caused by a protist (*Trypanosoma*).
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119. (3)
The prokaryotic bacteria also possess flagella but these are structurally different from that of the eukaryotic flagella. The electron microscopic study of a cilium or the flagellum show that they are covered with plasma membrane. Their core called the axoneme, possesses a number of microtubules running parallel to the long axis. The axoneme usually has nine doublets of radially arranged peripheral microtubules, and a pair of centrally located microtubules. Such an arrangement of axonemal microtubules is referred to as the 9+2 array. The central tubules are connected by bridges

and is also enclosed by a central sheath, which is connected to one of the tubules of each peripheral doublets by a radial spoke. Thus, there are nine radial spokes. The peripheral doublets are also interconnected by linkers. Both the cilium and flagellum emerge from centriole-like structure called the basal bodies.

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120. **(4)**

Solanaceae and Anacardiaceae are both families. Primata, Diptera and Poales are orders. Mammalia and Insecta are classes. Angiospermae is a division.

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121. **(4)**

S or synthesis phase marks the period during which DNA synthesis or replication takes place. During this time the amount of DNA per cell doubles. If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell had diploid or 2n number of chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 2n.

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122. **(1)**

The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

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123. **(2)**

The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. Each homologous chromosome consists of two sister chromatids, and each chromosome has one centromere.

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124. **(2)**

Most prokaryotic cells, particularly the bacterial cells, have a chemically complex cell envelope. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. The plasma membrane is composed of a phospholipid bilayer. The vacuole is bound by a single membrane called tonoplast. Animal cells contain another non-membrane bound organelle called centrosome which helps in cell division.

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125. **(3)**

The three-domain system has also been proposed that divides the Kingdom Monera into two domains, leaving the remaining eukaryotic kingdoms in the third domain and thereby a six kingdom classification.

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126. **(2)**

Synthesis of proteins required for DNA replication occurs during G_1 phase of interphase. Synthesis of copy of each chromosome happens during S phase of interphase. Formation of tetrad of chromosomes occurs during prophase I of meiosis I. Homologous chromosomes pair up to form tetrads (also called bivalents). This is where crossing over happens. Formation of dyad of cells happens after meiosis I. Formation of tetrad of cells happens after meiosis II.

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127. **(2)**

In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments. RER is frequently observed in the cells actively involved in protein synthesis and secretion. They are extensive and continuous with the outer membrane of the nucleus. The smooth endoplasmic reticulum is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesised in SER. Many membrane bound minute vesicles called microbodies that contain various enzymes, are present in both plant and animal cells.

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128. **(1)**

Mayr joined Harvard's Faculty of Arts and Sciences in 1953 and retired in 1975, assuming the title *Alexander Agassiz Professor of Zoology Emeritus*.

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129. **(1)**

The fungi constitute a unique kingdom of heterotrophic organisms. They show a great diversity in morphology and habitat. With the exception of yeasts which are unicellular, fungi are filamentous. The network of hyphae is known as mycelium. Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called coenocytic hyphae. They are cosmopolitan and are found in air, water, soil and on animals and plants.

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130. (1)

The two asters together with spindle fibres forms mitotic apparatus. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. An elaborate network of filamentous proteinaceous structures consisting of microtubules, microfilaments and intermediate filaments present in the cytoplasm is collectively referred to as the cytoskeleton. Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or polysome.

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131. (4)

Reproduction in fungi can take place by vegetative means – fragmentation, fission and budding. Asexual reproduction is by spores called conidia or sporangiospores or zoospores, and sexual reproduction is by oospores, ascospores and basidiospores. Sexual spores in sac fungi are called ascospores which are produced endogenously in sac like asci (singular ascus). In basidiomycetes, karyogamy and meiosis take place in the basidium producing four basidiospores. The basidiospores are exogenously produced on the basidium (pl.: basidia). E.g.; *Agaricus* (Mushroom). The deuteromycetes reproduce only by asexual spores known as conidia.

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132. (2)

Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

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133. (4)

Lichens are symbiotic associations i.e. mutually useful associations, between algae and fungi. The algal component is known as phycobiont and fungal component as mycobiont, which are autotrophic and heterotrophic, respectively. Algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner. So close is their association that if one saw a lichen in nature one would never imagine that they had two different organisms within them. Lichens are very good pollution indicators – they do not grow in polluted areas.

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134. (1)

The Golgi cisternae are concentrically arranged near the nucleus with distinct convex *cis* or the forming face and concave *trans* or the maturing face.

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135. (3)

The scope of systematics was later enlarged to include identification, nomenclature and classification. Systematics takes into account evolutionary relationships between organisms. The word systematics is derived from the Latin word ‘systema’ which means systematic arrangement of organisms.

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■■■

HINTS & SOLUTIONS

136. (1)

Human beings have many different tissues: epithelial, connective, muscle, nervous, etc. These tissues perform specialized functions.

All cells in the human body (except gametes) have the same genetic constitution (same DNA). Differences in function arise due to differential gene expression, not different genes.

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137. (2)

Organisms require energy to carry out various life processes such as movement, growth, and repair. This energy is derived by breaking down nutrient molecules, mainly glucose, through a process called cellular respiration. In aerobic respiration, oxygen does not directly break down glucose, but rather plays a critical role in the final step of the process that's done through enzymes and intermediate steps. Oxygen is essential in the ETC to keep the flow of electrons moving; without it, the entire chain would stop, halting ATP production.

Extracting nutrients from food is part of digestion, not respiration.

"Burn" implies direct combustion, which doesn't happen in cells. Cellular respiration is a controlled enzymatic process, not direct burning.

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138. (4)

The electrical activity of the heart is recorded using an electrocardiograph. The recording is called an electrocardiogram (ECG).

Heart attack: When the heart muscle is suddenly damaged by an inadequate blood supply.

Cardiac arrest: Heart suddenly stops beating. So, they are different.

Blood flow path: Aorta → Arteries → Arterioles → Capillaries → Tissues

Venules collect deoxygenated blood from tissues — not part of oxygenated delivery.

Hepatic portal circulation goes from digestive organs → liver → systemic venous system → heart → pulmonary circulation. So, blood does not return to pulmonary circulation directly after the liver.

The parasympathetic nervous system, via vagus nerve, releases acetylcholine. It slows the heart rate, reduces conduction velocity, and hence lowers cardiac output.

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139. (4)

Cockroaches are dioecious (i.e., male and female are separate individuals). Male and female cockroaches have distinct reproductive organs. Frogs have separate sexes (male and female frogs), so they are dioecious. *Rana tigrina* is unisexual i.e., individual organisms possess only one type of reproductive organ, either male or female.

Kidneys of frogs are compact, dark red and bean like structures situated a little posteriorly in the body cavity on both sides of the vertebral column.

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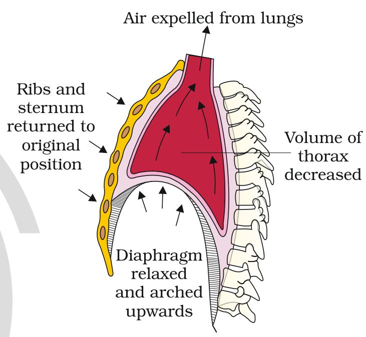
140. (2)



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141. (3)

List-I	List-II
Superior vena cava	Brings deoxygenated blood from upper part of body to right atrium
Inferior vena cava	Carries deoxygenated blood from lower body to right atrium
Pulmonary artery	Transports deoxygenated blood to the lungs
Pulmonary vein	Carries oxygenated blood to left atrium

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142. (3)

Cockroaches belong to phylum Arthropoda, class Insecta. *Periplaneta americana* are about 34–53mm. Wings extend beyond the abdomen in males. The dorsal plates are called tergites, not sternites. Sternites are on the ventral side.

Cockroaches have biting and chewing type mouthparts (mandibulate type).

Labrum = Upper lip

Labium = lower lip

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143. (3)

Only about 25% of oxygen carried by haemoglobin is delivered to tissues at rest. The remaining 75% stays bound to haemoglobin as a reserve. During heavy exercise, this extra oxygen is released to meet the high metabolic demand.

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144. (1)
In birds, the heart and circulatory system are highly adapted for sustained high metabolic demands (e.g., flight), including:
- Higher cardiac output
- Faster heartbeat
- Greater oxygen delivery efficiency

Thus, it's considered functionally superior or more efficient compared to reptiles.
A four-chambered heart completely separates:
- Oxygenated blood (from lungs) in the left side
- Deoxygenated blood (from body) in the right side

This allows for efficient oxygen delivery and high metabolic activity.
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145. (1)
Paramoecium is a unicellular protozoan. All life functions (movement, feeding, digestion, excretion, etc.) are performed by a single cell. Hence, no division of labour at tissue or organ level.
Hydra, *Taenia*, *Periplaneta* all are multicellular and hence show division of labour.
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146. (1)
Nearly 20-25 per cent of CO_2 is transported by RBCs whereas 70 per cent of it is carried as bicarbonate. About 7 per cent of CO_2 is carried in a dissolved state through plasma.
(NCERT Ed. 2025-26, 11th, Page No. 189)

147. (3)
The AV node introduces a slight delay in the transmission of the electrical impulse, allowing the atria to contract and push blood into the ventricles before the ventricles begin contracting. This coordination ensures efficient filling and pumping.
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148. (4)
The matrix of bone is hard and non-pliable due to calcium salts. The matrix of cartilage is solid and pliable, not hard due to hyaluronic acid and chondroitin salts.
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149. (2)
Carbon monoxide (CO) is a colorless, odorless, and toxic gas that poses a serious threat because:
- It binds very strongly with haemoglobin (Hb) to form carboxyhaemoglobin (HbCO).
- CO's affinity for haemoglobin is about 200–250 times higher than that of oxygen.
- This blocks oxygen from binding to haemoglobin, thereby reducing oxygen delivery to tissues.
- This leads to hypoxia (oxygen deficiency), especially affecting the brain and heart.

CO does not break down haemoglobin; it functionally disables it by binding strongly.
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150. (3)
The recipient with blood group A has anti-B antibodies in their plasma. If they receive blood from a group B donor, these antibodies will recognize B antigens on the donor RBCs and cause agglutination (clumping) — a potentially life-threatening transfusion reaction.
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151. (3)
Squamous epithelium has tightly packed polygonal cells (tile-like) that resemble a mosaic and hence called tessellated epithelium.
Like most epithelial tissues, squamous epithelium lacks blood vessels. In the alveoli of lungs, squamous epithelium forms the thin diffusion surface between blood and air.
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152. (2)

List-I	List-II
Inspiratory Capacity (IC)	Total air, a person can inspire after normal expiration.
Oxygen	Has no significant role in regulation of respiration.
Functional Residual Capacity (FRC)	Volume of the air that will remain in lungs after a normal expiration.
Spirometer	Cannot measure residual volume.

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153. (2)
Regular exercise strengthens the heart muscle, increasing stroke volume (amount of blood pumped per beat). As a result, the heart doesn't need to beat as frequently at rest, so resting heart rate decreases. Since cardiac output (CO = HR × SV) remains constant at rest, a higher stroke volume compensates for the lower heart rate. So, both statements — about unchanged cardiac output and increased stroke volume — are true.
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154. (2)

The primary function of the C-shaped (incomplete) rings of hyaline cartilage is to provide structural support and rigidity to the airways. This support prevents any collapse when the air pressure is low, such as during a forceful exhalation, ensuring the pathway for air remains open.

Lecithin is a type of phospholipid and a major component of pulmonary surfactant. Surfactant works by reducing surface tension inside the alveoli, preventing them from collapsing. It does not provide structural support to the bronchioles.

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155. (3)

The correct sequence for O_2 movement in the lungs, from the alveolar air into the blood, involves crossing several layers of the respiratory membrane. The path oxygen takes:

Alveolar epithelial cells (E): Oxygen first diffuses across the thin cells that form the wall of the air sac (alveolus).

Alveolar basement membrane (D): Next, it crosses the basement membrane that supports the alveolar epithelial cells.

Interstitial fluid (C): A tiny fluid-filled space between the alveolar membrane and the capillary membrane.

Capillary basement membrane (A): The basement membrane that supports the capillary endothelial cells.

Capillary endothelial cells (B): Finally, it crosses the cells that form the wall of the blood capillary to enter the plasma and then bind to haemoglobin within red blood cells.

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156. (2)

Erythroblastosis foetalis (also called haemolytic disease of the newborn) does not always occur in the first Rh^+ pregnancy of a Rh^- mother.

It usually occurs in subsequent pregnancies, if the mother becomes sensitized (i.e., produces anti-Rh antibodies) during a previous Rh^+ pregnancy.

Sensitization happens only when foetal blood leaks into the maternal circulation, which is more likely during delivery, miscarriage, or trauma.

Elevated WBC count is a typical immune response to infection, inflammation, or tissue damage. Leucocytosis (high WBC count) is a clinical indicator of infection in many conditions.

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157. (3)

The proximal convoluted tubule (PCT) of the nephron is lined by cuboidal epithelium with microvilli forming a brush border to increase surface area for reabsorption.

The intestinal lining, especially the small intestine, has simple columnar epithelium with microvilli (brush border) to increase surface area for absorption.

Keratinized epithelium is found on dry surfaces exposed to the environment (e.g., outer skin). Wet surfaces (like oral cavity, esophagus, vagina) have non-keratinized stratified squamous epithelium.

Ciliated columnar epithelium is present in:

- Fallopian tubes (oviducts) – help move the ovum.
- Bronchioles – helps in movement of mucus and trapped particles.

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158. (3)

The air present between nostrils and terminal bronchioles is termed as anatomical dead space, and it includes airways like the nostrils, pharynx, larynx, trachea, bronchi, and bronchioles (except respiratory bronchioles). No gaseous exchange occurs here because these regions lack alveoli. Only air that reaches the alveoli (in respiratory bronchioles, alveolar ducts, and alveoli) participates in gas exchange.

Residual volume is the air that remains in the lungs even after maximal exhalation. This prevents alveolar collapse by maintaining a certain level of pressure within the lungs, ensuring that alveoli stay partially filled with air. It works along with surfactant, which reduces surface tension, to help keep alveoli open.

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159. (1)

List I (Type of Leucocyte)	List II (Shape of Nucleus)
Neutrophil	Multilobed (3–5 lobes)
Lymphocyte	Spherical or round
Monocyte	Kidney-shaped
Eosinophil	Bilobed

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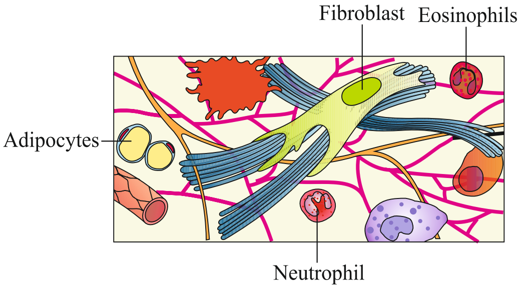
160. (1)



A- Adipocytes – Responsible for storage of fat
B- Neutrophil – Phagocytic action
C- Eosinophils – Responsible for allergic reactions
D- Fibroblast – Secretes collagen fibres

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161. (4)

Due to mixing with residual air and presence of water vapour and CO_2, partial pressure of O_2 in alveolar air is less than atmospheric air.
Atmospheric $pO_2 \approx 159$ mm Hg
Alveolar $pO_2 \approx 104$ mm Hg
Haemoglobin has a higher affinity for O_2 at the high pO_2 and low pCO_2 of alveoli.
Oxygen dissociation curve is sigmoid shaped due to cooperative binding of O_2 with haemoglobin.
Deoxygenated blood is more acidic than oxygenated blood because it contains more CO_2.
CO_2 forms carbonic acid (H_2CO_3) in blood, which dissociates into H^+ and HCO_3^-, increasing acidity.
Thus, deoxygenated blood has a lower pH than oxygenated blood.

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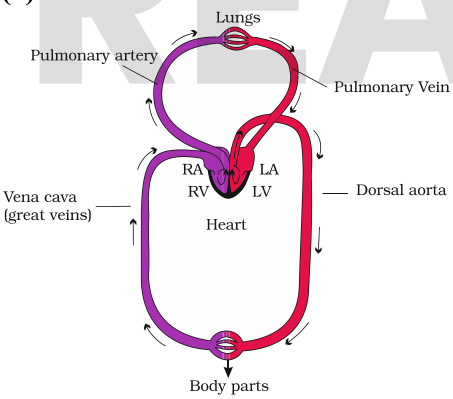
162. (2)



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163. (3)

Beheading a cockroach does not serve the purpose of killing it instantly. This is because the head holds a bit of the nervous system while the rest of the nervous system is situated along the ventral part of its body.

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164. (3)

A shift of the oxygen dissociation curve to the right facilitates the dissociation (release) of oxygen from haemoglobin, meaning haemoglobin's affinity for oxygen is decreased. This is beneficial for delivering more oxygen to the tissues, especially during strenuous exercise or other high metabolic states.
The factors that cause a rightward shift are:

- High pCO_2 (partial pressure of carbon dioxide): In the tissues, high levels of CO_2 lead to the formation of carbonic acid, which lowers pH and promotes oxygen release.
- High H^+ concentration (or low pH): An increase in acidity weakens the bond between haemoglobin and oxygen, causing oxygen to be released.
- Higher temperature: During physical activity, muscle temperature rises, which enhances the unloading of oxygen from haemoglobin to meet the increased metabolic demand.

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165. (1)

Lymph is a colorless (not red) fluid. It is low in large plasma proteins and does not contain RBCs or most formed elements. It contains some WBCs, especially lymphocytes, crucial for immune responses. Lymph is formed from interstitial (tissue) fluid that leaks out of blood capillaries. Lymph helps in the transport of nutrients, hormones, and waste materials from tissues to the bloodstream.
Dietary fats are absorbed as chylomicrons into lacteals (specialized lymph vessels in villi).

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166. (1)

Structure	Found in	Function
Anal cerci	Both sexes	Paired, short, thread-like appendages from the 10th segment, used for sensory perception.
Anal styles	Males only	Slender, unsegmented, jointed structures, also from the 9th sternum, help during copulation.

Cockroach A has:

- Cerci (common to both sexes)
- Styles → present only in males, So, A = Male
- Cockroach B has:
  - Only cerci but no styles, So, B = Female

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167. (3)

When a person ascends to high altitudes (e.g., > 3,500 meters), they often experience:

- Breathlessness (dyspnea)
- Fatigue

This is due to hypoxia, where:

- The atmospheric pressure drops at high altitudes.
- As a result, inspired pO_2 and consequently alveolar and arterial pO_2 fall.
- Even though lungs and haemoglobin are normal, less oxygen is available for gas exchange → reduced oxygen supply to tissues.

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168. (4)

P-wave → Atrial depolarisation

QRS complex → Ventricular depolarisation (and atrial repolarisation, though it's hidden within the complex)

T-wave → Ventricular repolarisation

U-wave → Comes after T-wave

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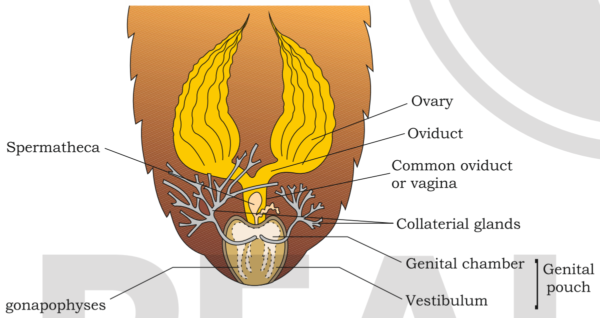
169. (3)



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170. (4)

Stretch receptors in the lungs send inhibitory signals to the respiratory centers to prevent over-inflation during forced inspiration.

Cigarette smoke contains carcinogens like benzopyrene, which damage the lung tissue and can lead to lung cancer. They damage the alveolar wall and decrease exchange surface area leading to emphysema.

The pneumotaxic centre (in the pons) limits the duration of inspiration. It sends inhibitory signals to the inspiratory centre (medulla), thus reducing inspiration time and increasing breathing rate.

External intercostal muscles lift up the ribs and sternum during inspiration. Internal intercostal muscles are used during forced expiration; they pull the ribs downward and inward.

In certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long exposure can give rise to inflammation leading to fibrosis (proliferation of fibrous tissues) and thus causing serious lung damage.

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171. (3)

The cardiac cycle starts with joint diastole (all chambers relaxed).

I. Joint diastole (Image 2): Both atria and ventricles are relaxed and filling.

II. Atrial systole (Image 1): Atria contract, pushing blood into the ventricles.

III. Ventricular systole (Image 3): Ventricles contract, pumping blood out.

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172. (2)

List-I	List-II
Mushroom gland	6th -7th segment
Abdominal ganglion	6 in number
Phallomeres	9th sternum
Total abdominal segments in male	10 segments

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173. (1)

Trachea and primary bronchi have C-shaped (dorsally incomplete) cartilaginous rings.

At high altitude, atmospheric pO_2 is low, causing hypoxia. In response, kidneys release erythropoietin, stimulating RBCs production (a process called acclimatization).

The left lung has two lobes and features a cardiac notch to accommodate the heart.

This is an anatomical hallmark of the left lung.

Normal expiration is passive due to lung recoil.

But forceful expiration involves active contraction of:

- Internal intercostal muscles
- Abdominal muscles

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174. (2)

The heart sounds "lub" and "dub" are associated with specific events in the cardiac cycle:

- "Lub" → Closure of atrioventricular (AV) valves (tricuspid and bicuspid) at the beginning of ventricular systole.
- "Dub" → Closure of semilunar valves (aortic and pulmonary) at the end of ventricular systole and the beginning of ventricular diastole.

Cardiac Cycle Timing Overview (Avg 0.8 sec total):

- Atrial systole: ~0.1 sec
- Ventricular systole: ~0.3 sec
- Joint diastole (relaxation phase): ~0.4 sec

The "lub" to "dub" interval corresponds to the ventricular systole, which is about 0.3 seconds.

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175. (2)

Grinding of food particles is done by the gizzard (also called proventriculus) which is located between the crop and midgut and has chitinous plates for mechanical grinding.
Secretion of digestive juices is done by the gastric caeca located at the junction of foregut and midgut.
Clearing of haemolymph is done by Malpighian tubules that extract nitrogenous waste from haemolymph.

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176. (2)

CO_2 dissolves in blood and cerebrospinal fluid (CSF) to form carbonic acid (H_2CO_3), which dissociates into H^+ and HCO_3^-:
$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$
This increase in H^+ ions lower the pH of the blood and CSF.
The medulla oblongata (specifically chemosensitive area) does not directly detect CO_2, but it detects the pH of the cerebrospinal fluid (CSF). When pH drops (due to high CO_2), the medulla increases the rate and depth of breathing → more CO_2 is expelled → pH is restored.

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177. (4)

Lung congestion (fluid accumulation in the lungs) is a classic sign of left ventricular heart failure. When the left side of the heart fails to pump blood efficiently, blood backs up into the pulmonary circulation, causing:

- Pulmonary congestion
- Shortness of breath
- Fatigue

This is a key feature of congestive heart failure, not necessarily of a heart attack, cardiac arrest, or myocardial infarction or SA node failure although they may be related.

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178. (3)

Vasa efferentia are 10-12 in number that arise from testes. They enter the kidneys on their side and open into Bidder's canal. Finally, it communicates with the urinogenital duct that comes out of the kidneys and opens into the cloaca.

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179. (4)

On an average, a healthy human breathes 12-16 times/minute.
The volume of air involved in breathing movements can be estimated by using a spirometer which helps in clinical assessment of pulmonary functions.
During inspiration, the diaphragm contracts and moves downward, expanding thoracic volume and drawing air in. During expiration, the diaphragm relaxes and returns upward, reducing thoracic volume and pushing air out. While expiration is passive in normal breathing, the diaphragm's relaxation is still essential.

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180. (3)

The CO_2 originates in a systemic arteriole (right thumb).
It enters the first capillary bed in the thumb (where CO_2 diffuses from tissues into blood).
It travels via veins to the right side of the heart, then to the lungs.
In the lungs, it passes through a second capillary bed (pulmonary capillaries) where CO_2 diffuses into the alveoli to be exhaled.
Hence, minimum two capillary beds are involved:

- Systemic capillary in the thumb
- Pulmonary capillary in the lungs

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